

Implementing solutions: A review exploring determinants for the implementation of solutions to food-energy-water system challenges

Supplemental materials for submittal to Frontiers in Environmental Science

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Supplemental Materials Summary

This analysis focuses on examining if how stakeholder engagement, and the level of engagement, impacts whether a solution for research outcomes is proposed and/or implemented. This meta-synthesis of 483 papers were evaluated and coded using several differing engagement scales. Additionally, each paper was coded by the geographic scale, and whether a computational model was used as part of the research.

Variable Summary

Below is a list of the categorical variables generated from the literature reviews

Table 1: Table T1: Variable Descriptions

Variable Name	Description
Year	Year of citation
Solution Proposed	Was a solution proposed?
Solution Implemented	Was a solution implemented?
Solution Type	If a solution was proposed, what was the solution type? Groups include: Technology, Policy, Institutional, Social, Economic, Ecological, and Educational.
Researcher Type	What was the research type? Groups include: NGO, English, Math, Computer Science, Physics, Engineering, Interdisciplinary, Social Science, Economics, Agriculture, and Other
Stakeholder Type	What was the stakeholder type? Groups include: Farmers, Combined Government, Combined Coalition, Combined Industry, Migrants, Youth, Public, Univerity, and Experts
Stakeholder engagement Scale - Ghodsvali	If a stakeholder was engaged, categorization of the engagement using the Ghodsvali scale. Groups include: Nominal, Instrumental, Representation, and Transformative
Geographical Type	What the geography type? Groups include: Not Described, Local, Regional, National, Multinational, Global, and No Geography
Region	What was the country?

Chi-Square Testing

Chi Square Testing: solution proposed or not vs. stakeholder engagement

3 Chi Square and Fishers Exact Test on contingency table with Solution/No Solution as the explanatory variable, and engaged stakeholder/did not engage stakeholder as the response variable.

ChiSquare = 46: Fishers Exact Test Odds Ratio: 17: Not Independent

Both chi square and fishers exact test were significant, with a chi square approximation of ~43, which is well above the critical value (3.84 with one degree of freedom). Fishers Exact Test returned an odds ratio of ~17. The alternative hypothesis: true odds ratio is not equal to 1, therefore the null hypothesis is rejected - the groups are not independent.

The Fishers Exact Test defaults to associating the odds ratio (which can represent effect size) with the first cell. In this instance “The odds of having a solution is 17 times that for an engaged stakeholder”. You could flip the response and explanatory variables, but the odds ratio would stay the same.

For more info on this topic see: Kim HY. Statistical notes for clinical researchers: Chi-squared test and Fisher’s exact test. Restor Dent Endod. 2017 May;42(2):152-155. doi: 10.5395/rde.2017.42.2.152. Epub 2017 Mar 30. PMID: 28503482; PMCID: PMC5426219.

```
## Loading required package: grid

##           stakeholder
## solution    N      Y
##           N 389   76
##           Y   4   14

## Number of cases in table: 483
## Number of factors: 2
## Test for independence of all factors:
##  Chisq = 43.14, df = 1, p-value = 5.104e-11
##  Chi-squared approximation may be incorrect

##
##  Fisher’s Exact Test for Count Data
##
## data:  solution_stakeholder
## p-value = 4.019e-08
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##   5.385841 76.064980
## sample estimates:
## odds ratio
##   17.75835

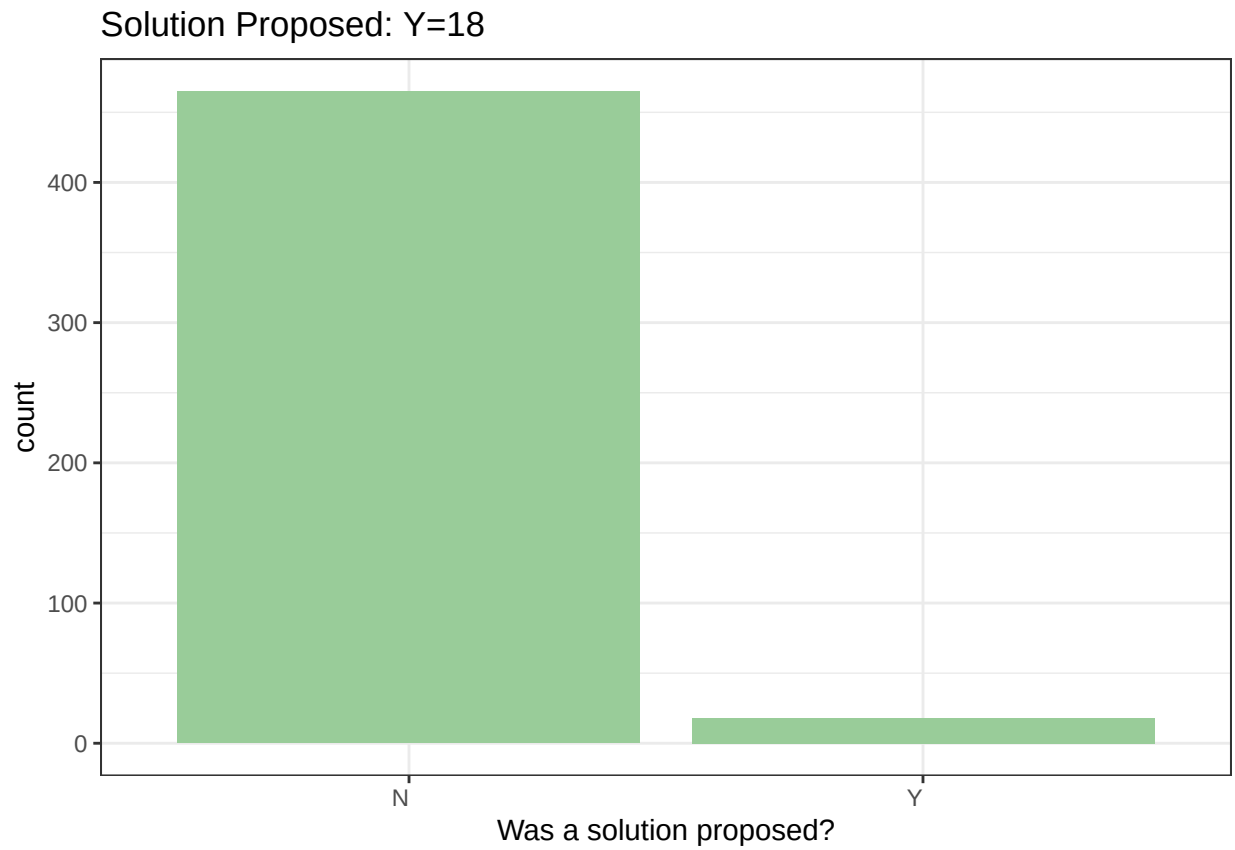
##
## Barnard’s Unconditional Test
##
##           Treatment I Treatment II
## Outcome I           76          465
## Outcome II          14           18
##
```

```
## Null hypothesis: Treatments have no effect on the outcomes
## Score statistic = 4.48684
## Nuisance parameter = 0.99 (One sided), 0.01 (Two sided)
## P-value = 0.000218524 (One sided), 0.000218524 (Two sided)
```

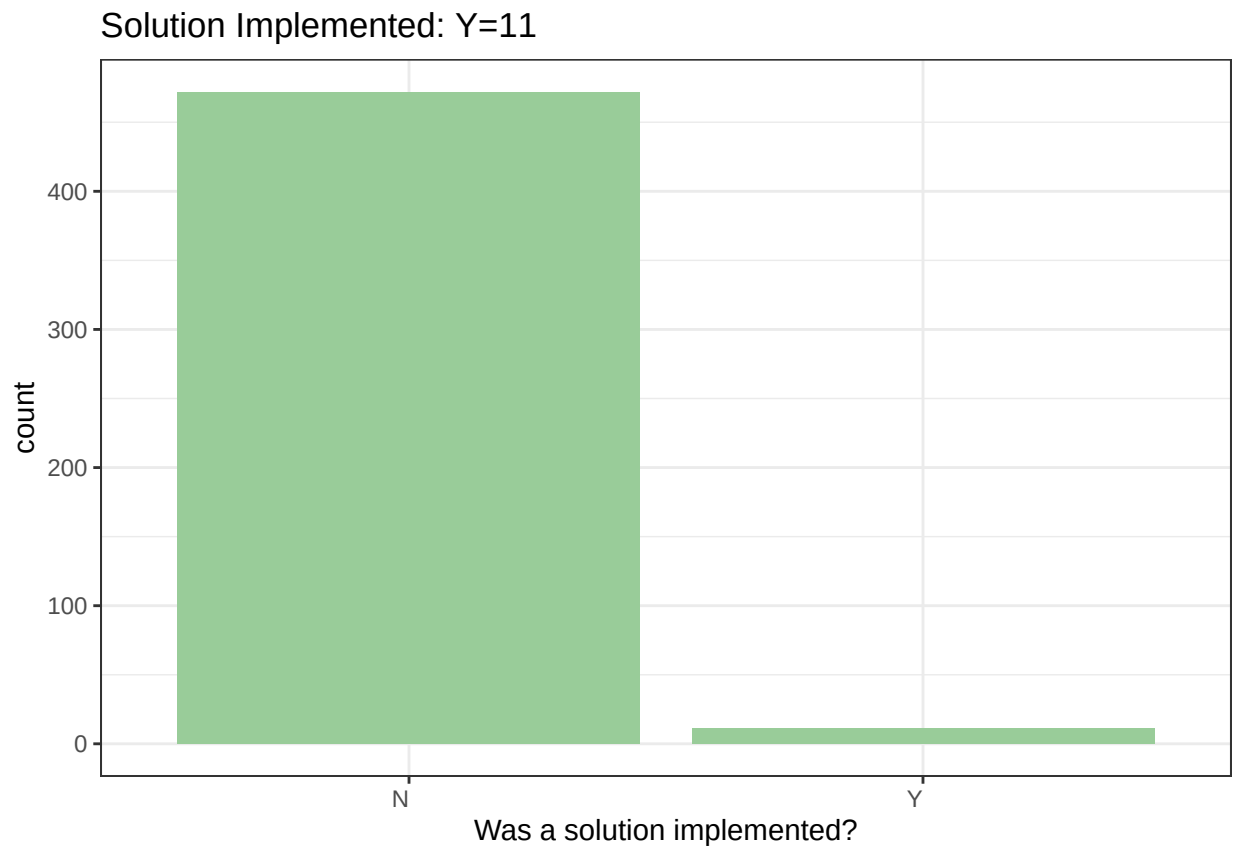
```
##              X^2 df    P(> X^2)
## Likelihood Ratio 31.288  1 2.2245e-08
## Pearson          43.137  1 5.1036e-11
##
## Phi-Coefficient   : 0.299
## Contingency Coeff.: 0.286
## Cramer's V        : 0.299
```

Summary Statistics Graphs

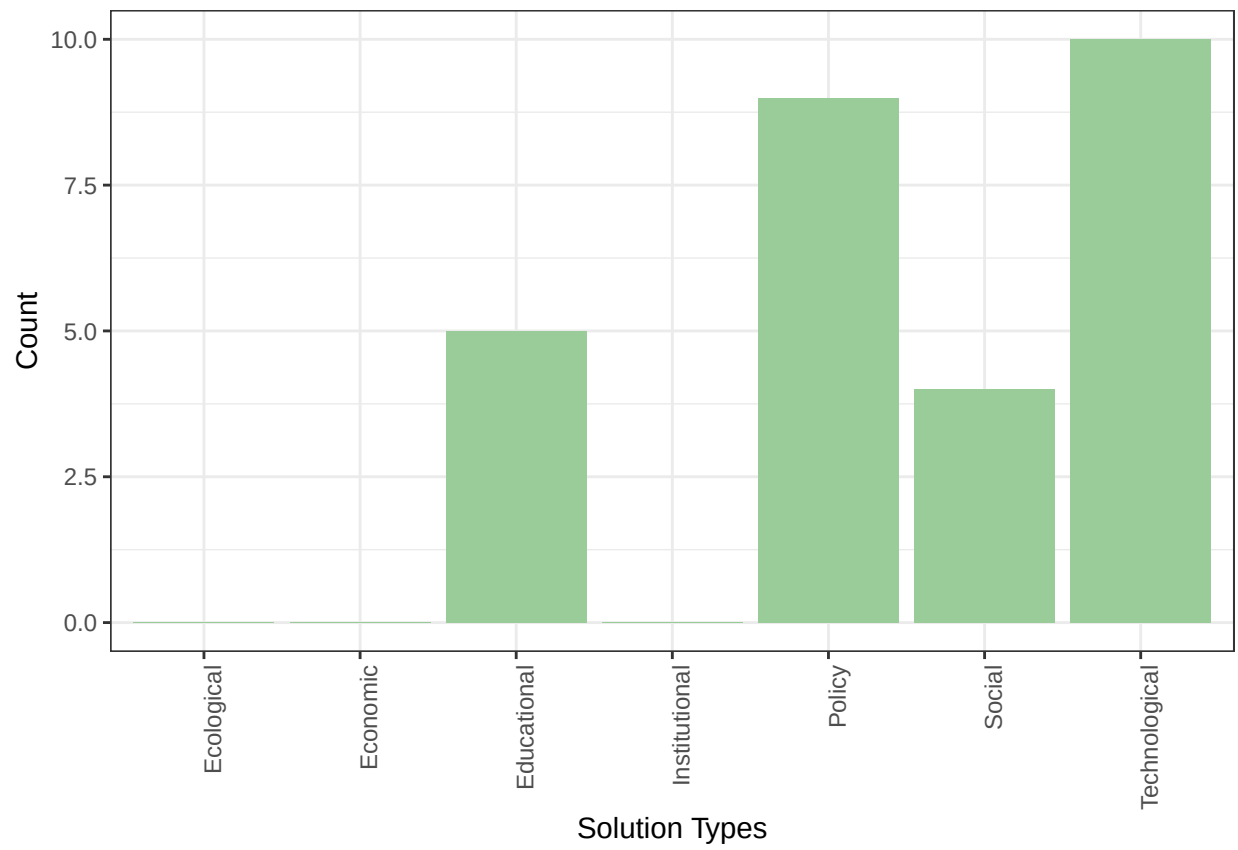
Were solutions proposed in the set of all papers?



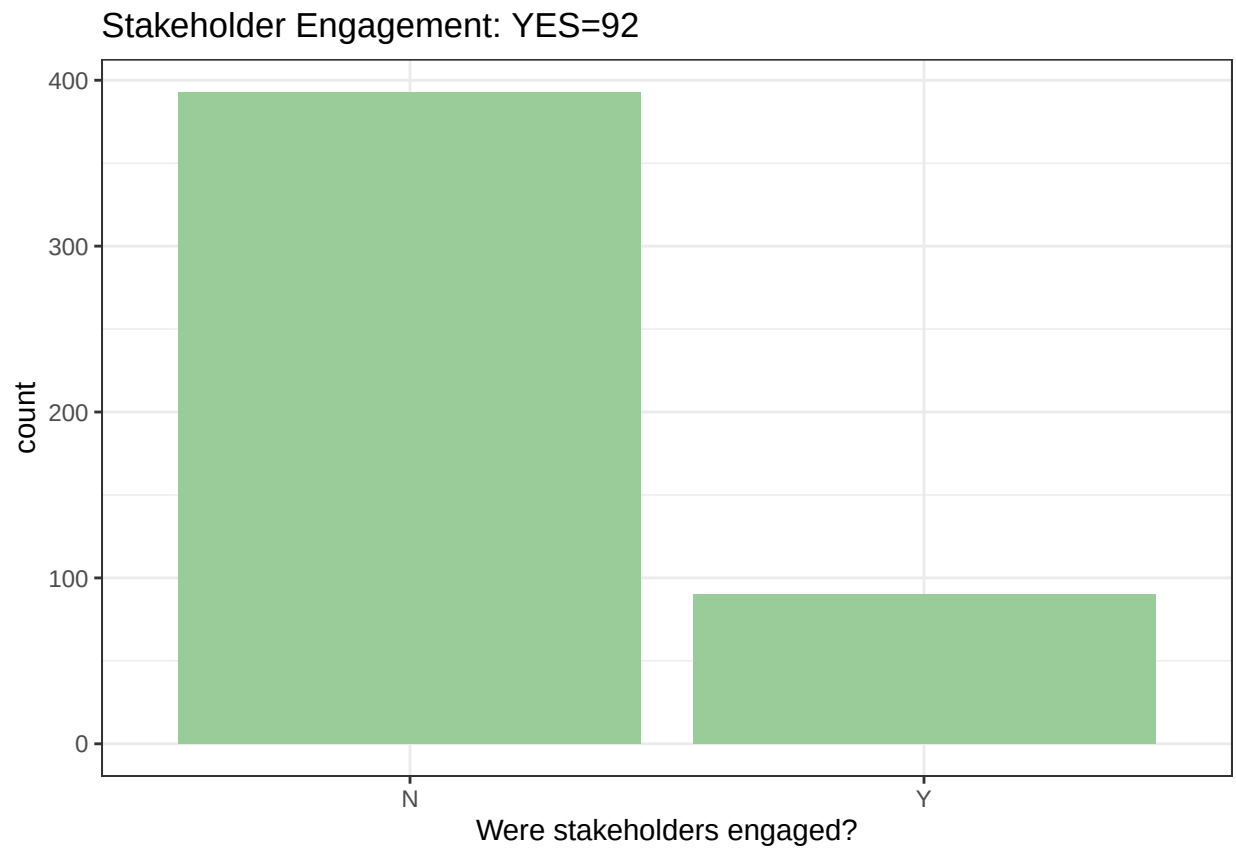
Were solutions implemented in the set of all papers?



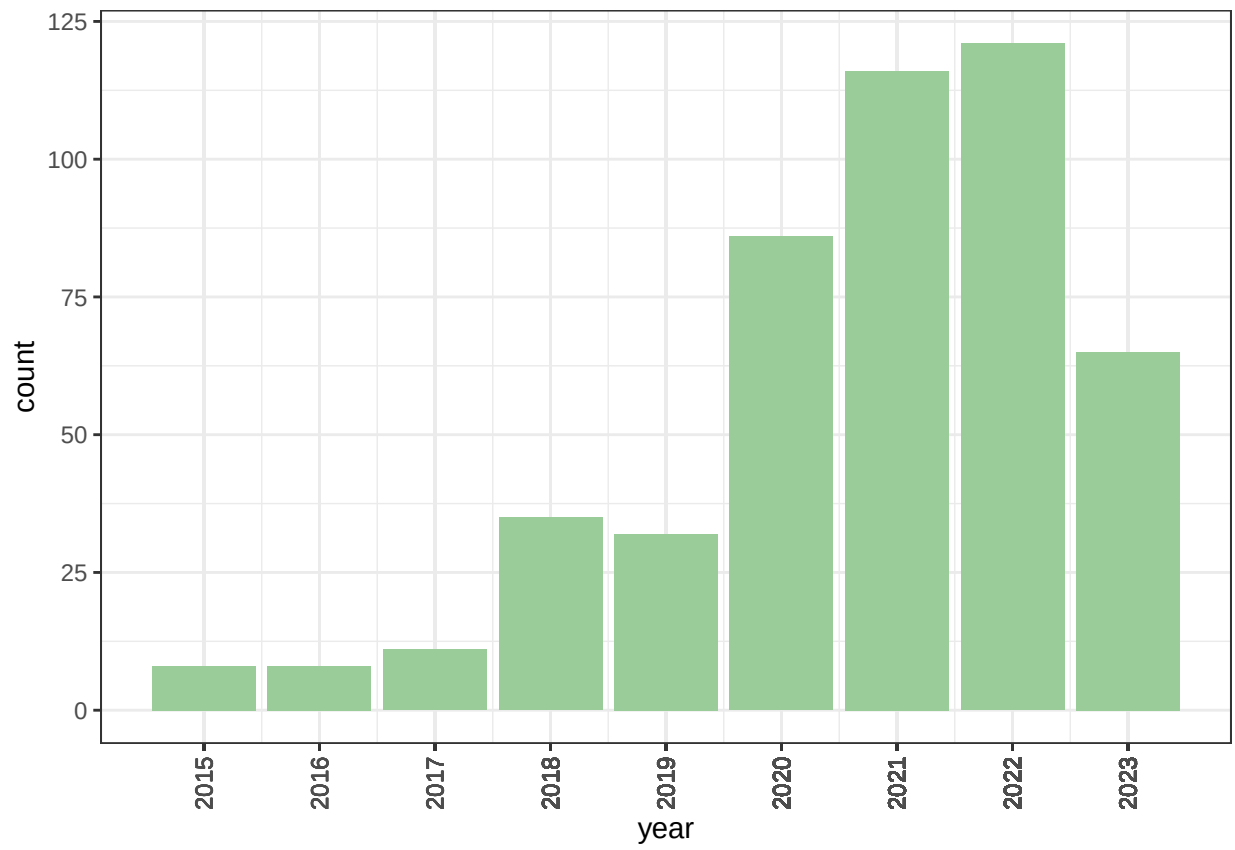
What were the solution types?



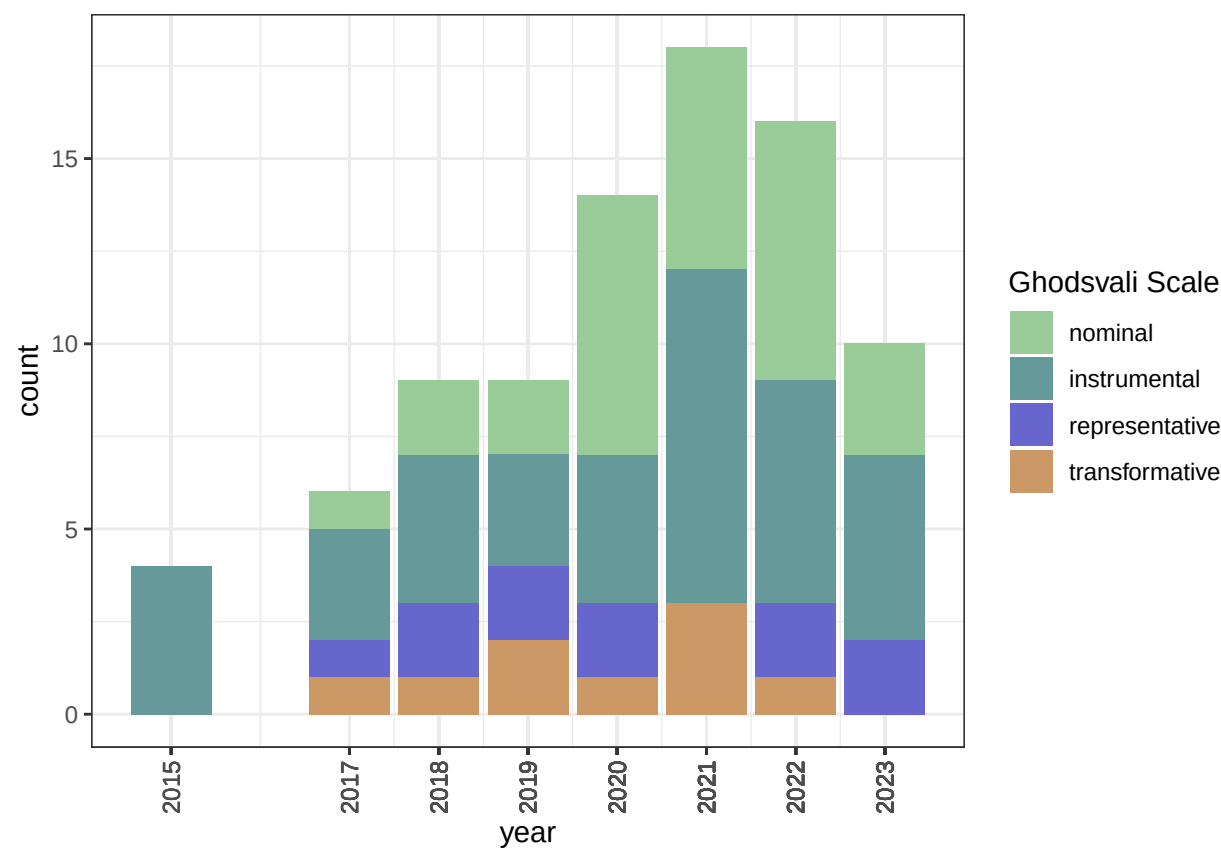
Were stakeholders engaged?



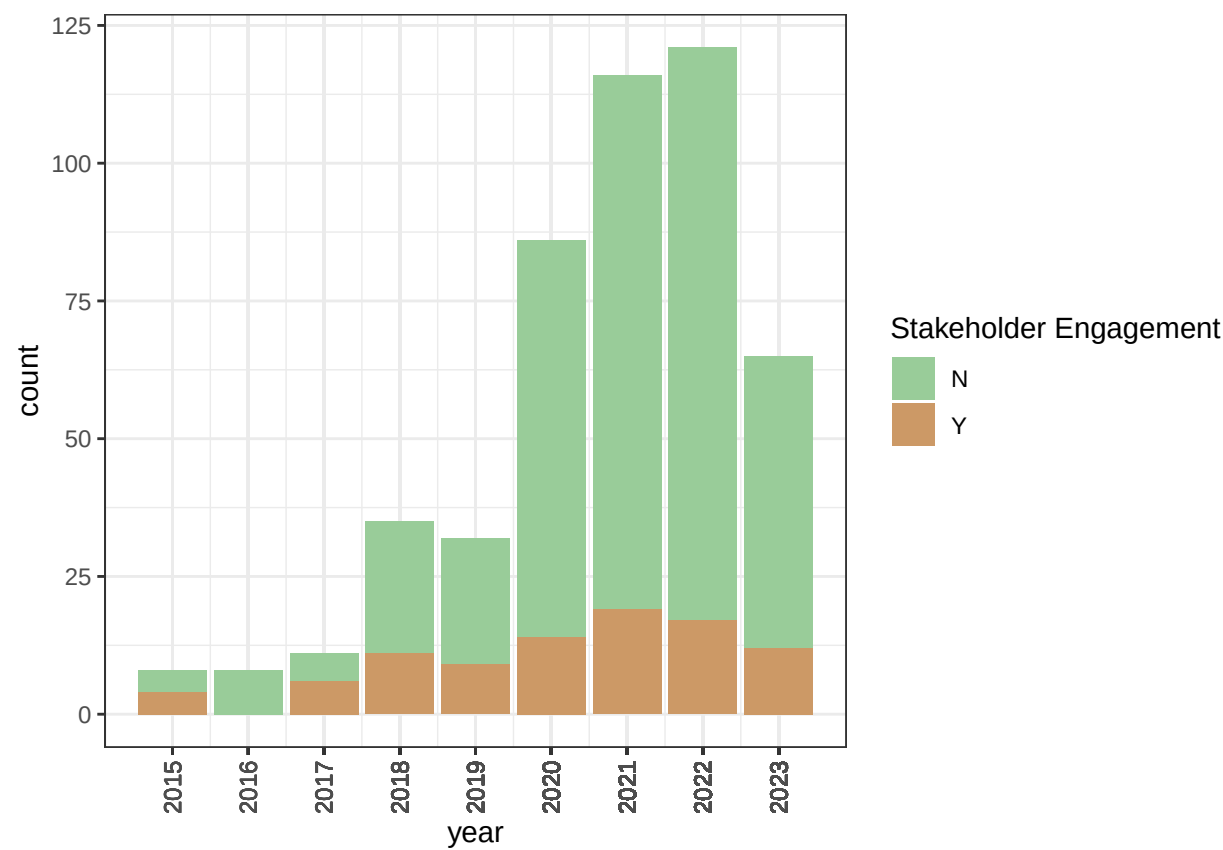
All FEWS papers by year



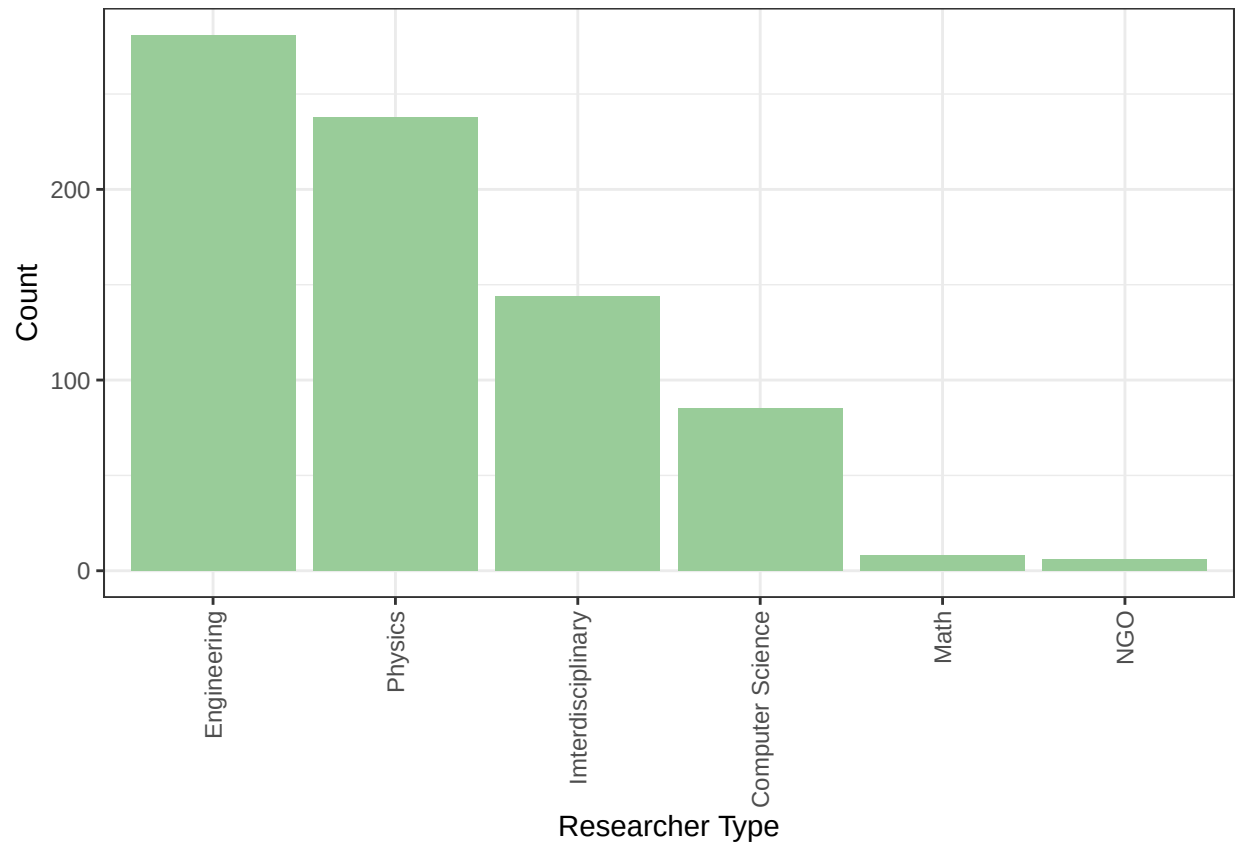
Level of stakeholder engagement by year - Ghodsvali scale



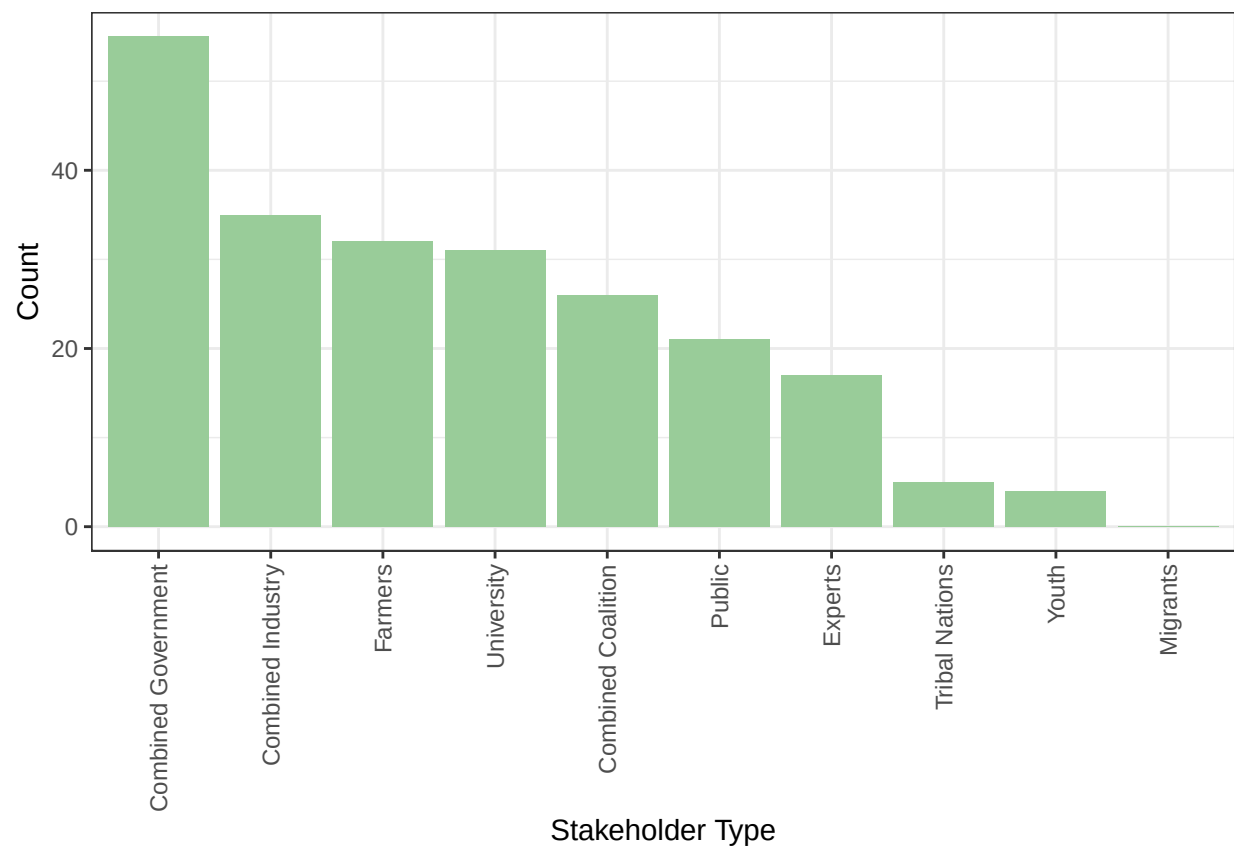
Stakeholder engagement by year



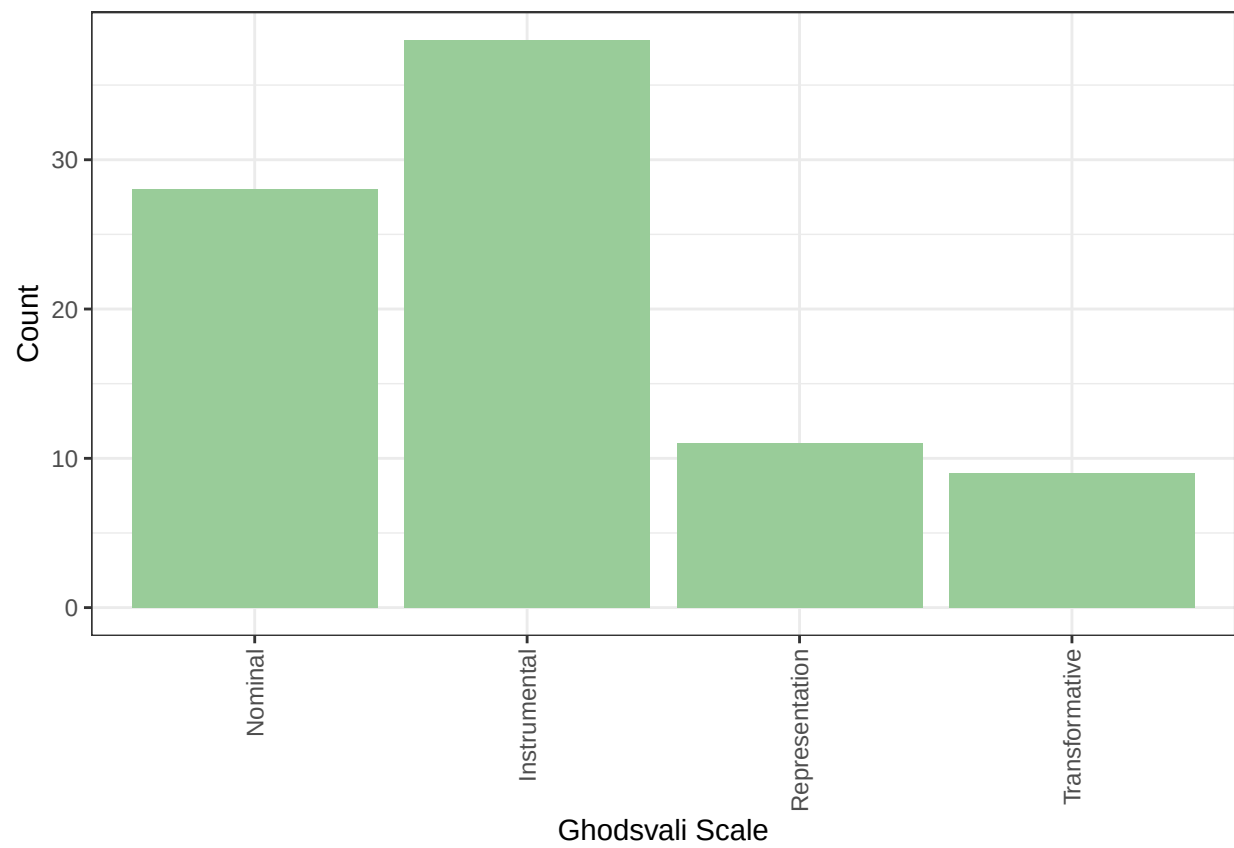
Researcher types



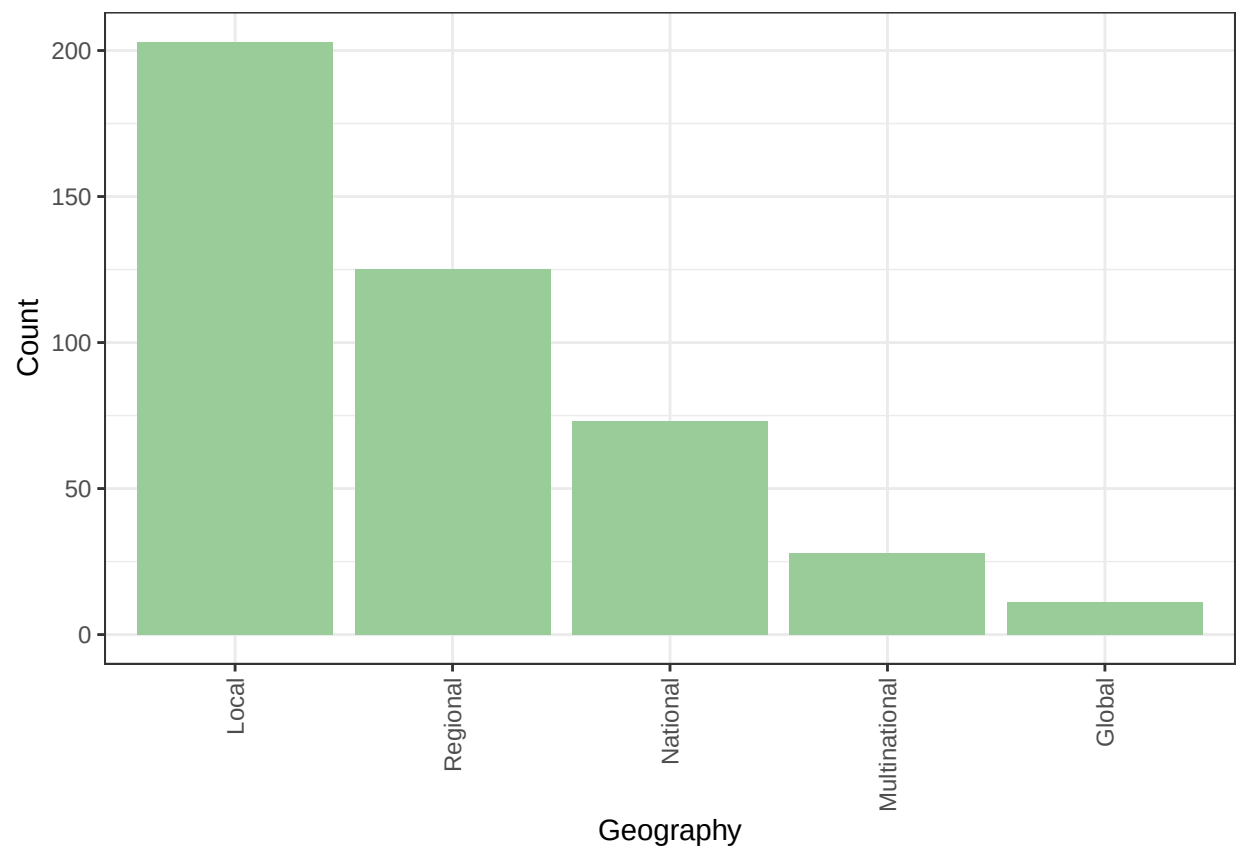
Stakeholder types



Ghodsvali scale breakdown



Geographic location breakdown



Ghodsvali Scale Modeling - solution proposed

Ghodsvali scale regression

Ghodsvali scale regression testing on whether a solution was proposed or not

```
##
## Call:
## glm(formula = solution_proposed_YN ~ STE_G_nominal + STE_G_instrumental +
##      STE_G_representation + STE_G_transformative, family = binomial(link = "logit"),
##      data = crcdata)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -4.5875     0.5025  -9.129  < 2e-16 ***
## STE_G_nominal     1.2917     1.1356   1.137  0.25535
## STE_G_instrumental 2.1308     0.7839   2.718  0.00656 **
## STE_G_representation 3.6067     0.8431   4.278 1.89e-05 ***
## STE_G_transformative 5.8403     0.9463   6.172 6.74e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 153.748  on 482  degrees of freedom
## Residual deviance:  96.785  on 478  degrees of freedom
## AIC: 106.79
##
## Number of Fisher Scoring iterations: 7
```

Ghodsvali scale odds

Odds of Ghodsvali scale predicting whether a solution was proposed or not

```
##
## Logistic regression predicting solution_proposed_YN : Y vs N
##
##               crude OR(95%CI)         adj. OR(95%CI)         P(Wald's test) P(LR-test)
## STE_G_nominal: 1 vs 0           0.95 (0.12,7.44)         3.64 (0.39,33.7)         0.255           0.318
##
## STE_G_instrumental: 1 vs 0       2.46 (0.68,8.9)          8.42 (1.81,39.14)         0.007           0.015
##
## STE_G_representation: 1 vs 0     11.42 (2.75,47.41)        36.84 (7.06,192.33)       < 0.001         < 0.001
##
## STE_G_transformative: 1 vs 0     147.32 (27.42,791.53)    343.87 (53.82,2197.12)   < 0.001         < 0.001
##
## Log-likelihood = -48.3926
## No. of observations = 483
## AIC value = 106.7851
```

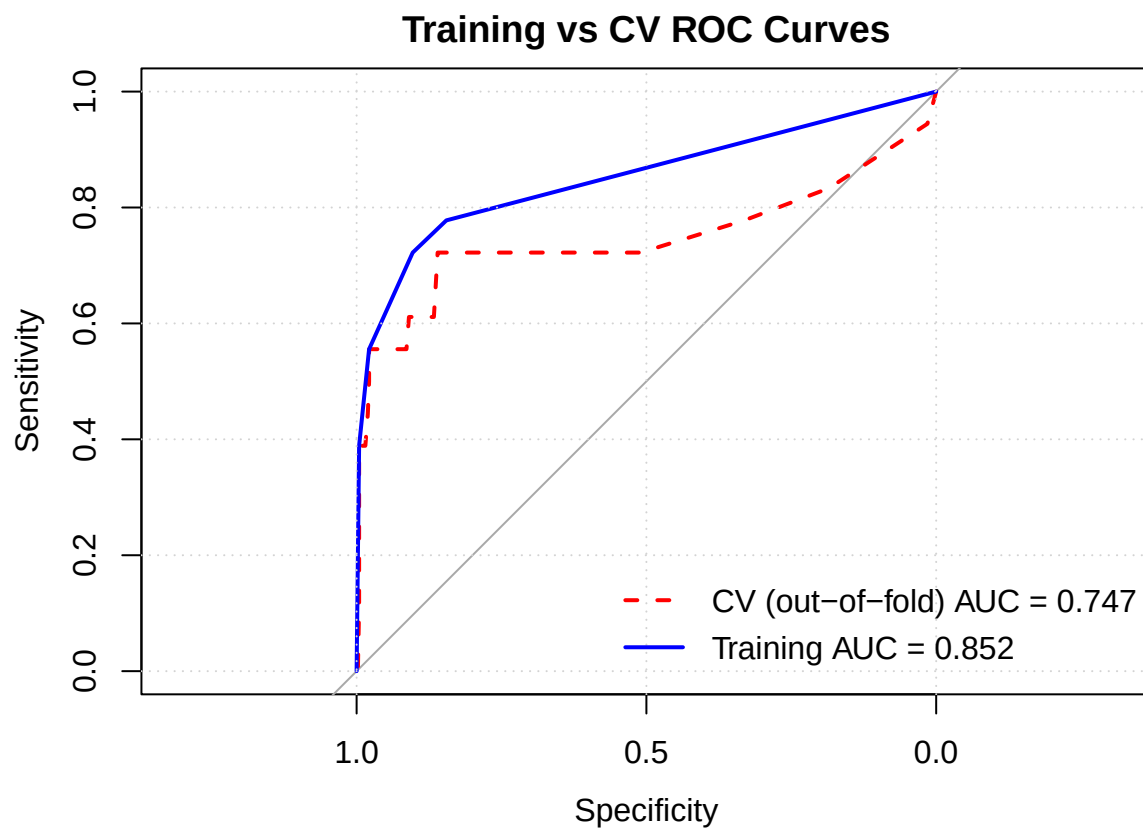
Ghodsvali Ensembled Decision Tree with Feature Importance

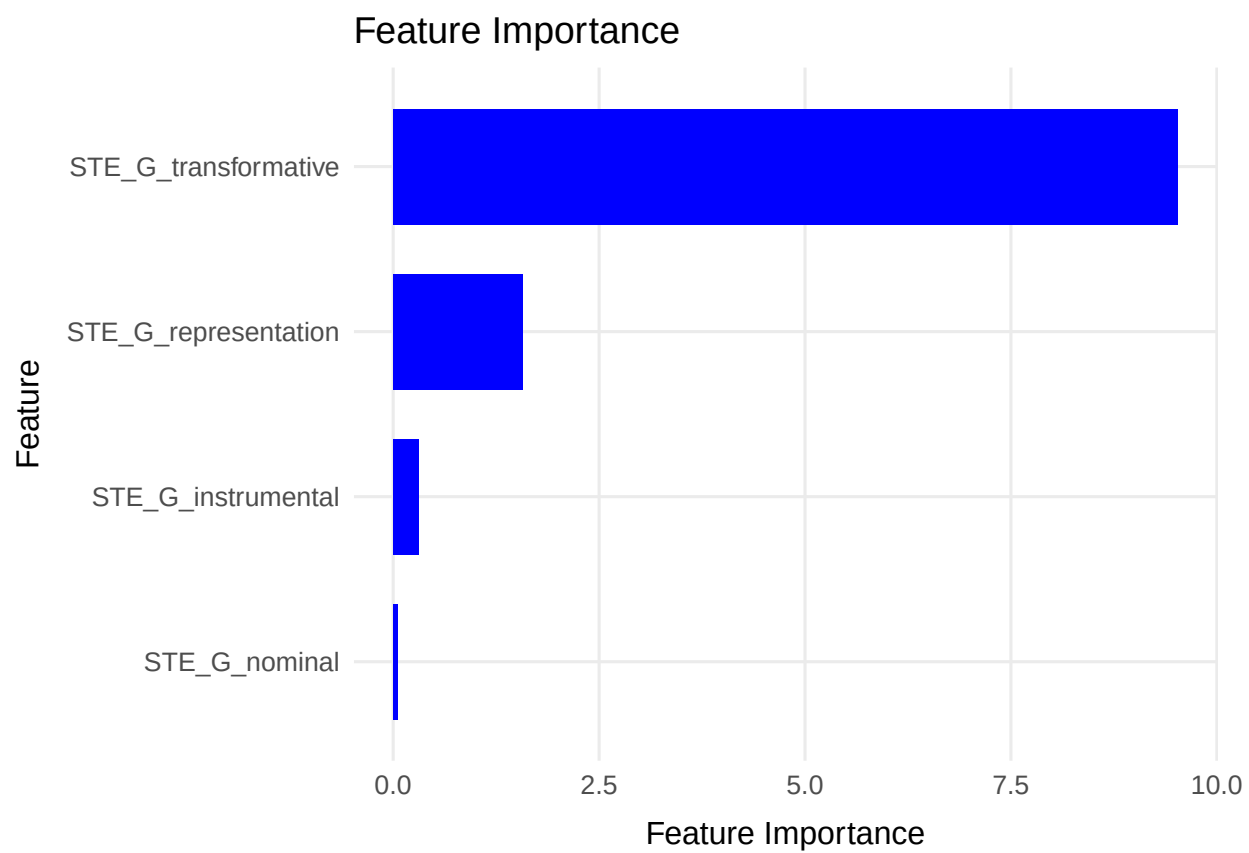
Ghodsvali ensembled decision tree with solution proposed being the dependent variable

```
## **Random forest (ranger).** 1000 trees; mtry=2; min.node.size=20; 5-fold CV.
```

```
## **CV AUC:** 0.818 (SD=0.197).  **n:** 483.  **Class counts:** N=465, Y=18.
```

```
## **Top features:** STE_G_transformative, STE_G_representation, STE_G_instrumental.
```





Stakeholder Engagement Modeling - solution proposed

REGRESSION: Does engaging stakeholders increase the likelihood that a solution will be proposed/implemented?

Here we use classical logistic regression using a binomial function to determine if engaging stakeholders (Y/N) increases the odds that a solution will be proposed.

```
##
## Call:
## glm(formula = solution_proposed_YN ~ S_stakeholder_engagement_YN,
##      family = binomial, data = crcdata)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -4.5773     0.5026  -9.108  < 2e-16 ***
## S_stakeholder_engagement_YNY  2.8856     0.5806   4.970 6.71e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 153.75  on 482  degrees of freedom
## Residual deviance: 122.46  on 481  degrees of freedom
## AIC: 126.46
##
## Number of Fisher Scoring iterations: 7
```

ODDS RATIOS: Does engaging stakeholders increase the likelihood that a solution will be proposed/implemented?

Odds of whether engaging stakeholders increases the likelihood that a solution will be proposed?

```
##
## Logistic regression predicting solution_proposed_YN : Y vs N
##
##                               OR(95%CI)          P(Wald's test) P(LR-test)
## S_stakeholder_engagement_YN: Y vs N 17.91 (5.74,55.91) < 0.001      < 0.001
##
## Log-likelihood = -61.23
## No. of observations = 483
## AIC value = 126.4599
```


Diversity of stakeholders vs solution

REGRESSION: Does the diversity of stakeholders increase the likelihood that a solution will be proposed?

Regression testing of whether Diversity of stakeholders predicts if a solution was proposed (Y/N). In order to represent diversity, we have used a simple ratio calculation which sums the number of stakeholders involved divided by the total number of possible stakeholder options. A ratio which is closer to 1 has a greater level of stakeholder diversity.

```
##
## Call:
## glm(formula = solution_proposed_YN ~ ST_ratio, family = binomial,
##      data = crcdata)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -4.0385      0.3569 -11.314  < 2e-16 ***
## ST_ratio       6.7128      1.2280   5.466  4.6e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 153.75  on 482  degrees of freedom
## Residual deviance: 126.74  on 481  degrees of freedom
## AIC: 130.74
##
## Number of Fisher Scoring iterations: 6
```

ODDS RATIOS: Does the diversity of stakeholders increase the likelihood that a solution will be proposed?

```
##
## Logistic regression predicting solution_proposed_YN : Y vs N
##
##               OR(95%CI)               P(Wald's test) P(LR-test)
## ST_ratio (cont. var.) 822.84 (74.13,9133.12) < 0.001      < 0.001
##
## Log-likelihood = -63.371
## No. of observations = 483
## AIC value = 130.742
```

REGRESSION: If diversity of stakeholders does not increase proposing/implementing solutions, which stakeholders are more associated with proposing/implementing solutions?

Regression testing for which stakeholders predict whether a solution was proposed

```
##
## Call:
## glm(formula = solution_proposed_YN ~ ST_farmers + ST_combined_gov +
##      ST_combined_coalition + ST_combined_industry + ST_public +
##      ST_university + ST_experts, family = binomial, data = crcdata)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -4.6904     0.5050  -9.287  < 2e-16 ***
## ST_farmers       0.5870     0.7238   0.811    0.417
## ST_combined_gov  4.0098     0.7989   5.019 5.18e-07 ***
## ST_combined_coalition -0.1027    0.8789  -0.117    0.907
## ST_combined_industry -1.0527    0.7426  -1.418    0.156
## ST_public        0.7677     0.7908   0.971    0.332
## ST_university   -0.2103     0.8365  -0.251    0.802
## ST_experts      -0.4590     0.7884  -0.582    0.560
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 153.75  on 482  degrees of freedom
## Residual deviance: 103.49  on 475  degrees of freedom
## AIC: 119.49
##
## Number of Fisher Scoring iterations: 7
```

ODDS RATIOS: stakeholders vs solution

Odds whether a specific stakeholder predicts if a solution was proposed (Y/N).

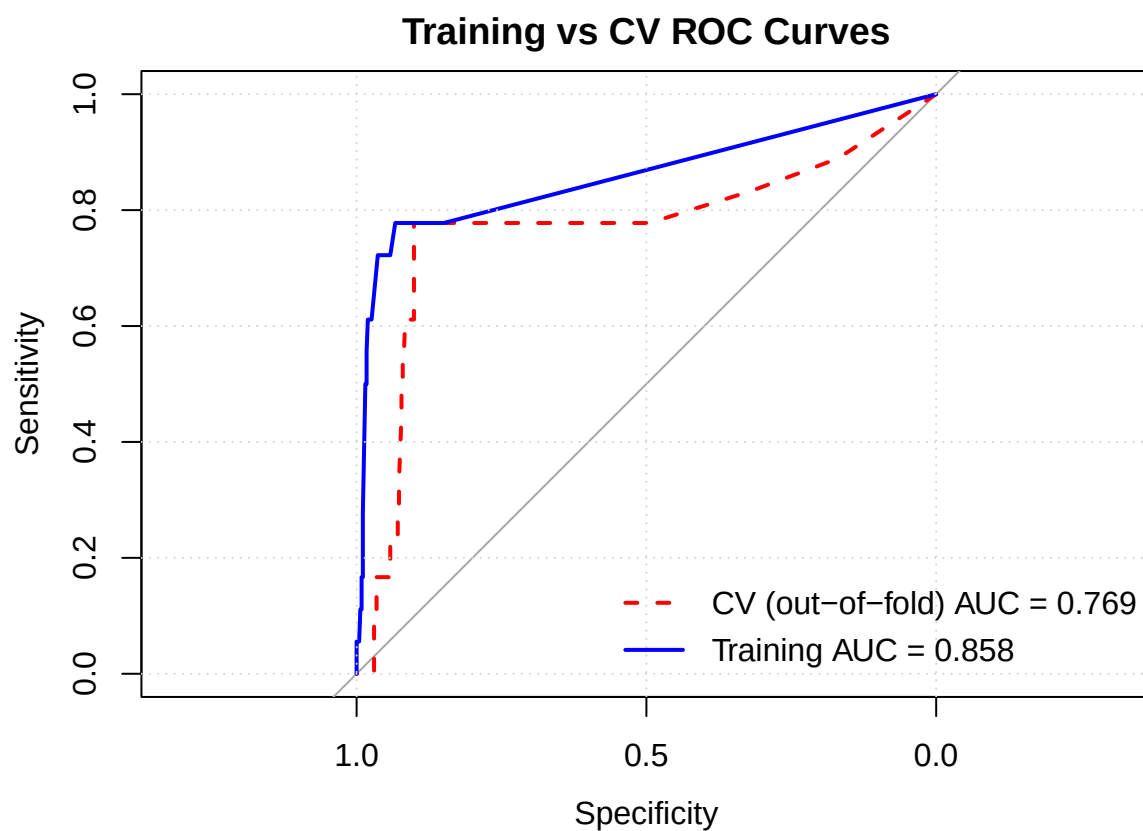
```
##
## Logistic regression predicting solution_proposed_YN : Y vs N
##
##               crude OR(95%CI)      adj. OR(95%CI)      P(Wald's test) P(LR-test)
## ST_farmers: 1 vs 0      6.24 (2.07,18.79)      1.8 (0.44,7.43)      0.417      0.42
##
## ST_combined_gov: 1 vs 0      36.2 (11.39,115.06)      55.13 (11.52,263.88) < 0.001      < 0.001
##
## ST_combined_coalition: 1 vs 0 8.13 (2.65,24.93)      0.9 (0.16,5.05)      0.907      0.907
##
## ST_combined_industry: 1 vs 0  5.58 (1.86,16.68)      0.35 (0.08,1.5)      0.156      0.144
##
## ST_public: 1 vs 0          9.93 (2.87,34.34)      2.15 (0.46,10.15)      0.332      0.34
##
## ST_university: 1 vs 0       8.8 (3.05,25.39)      0.81 (0.16,4.18)      0.802      0.801
##
## ST_experts: 1 vs 0         4.97 (1.32,18.71)      0.63 (0.13,2.96)      0.56      0.552
##
## Log-likelihood = -51.7449
## No. of observations = 483
## AIC value = 119.4899
```

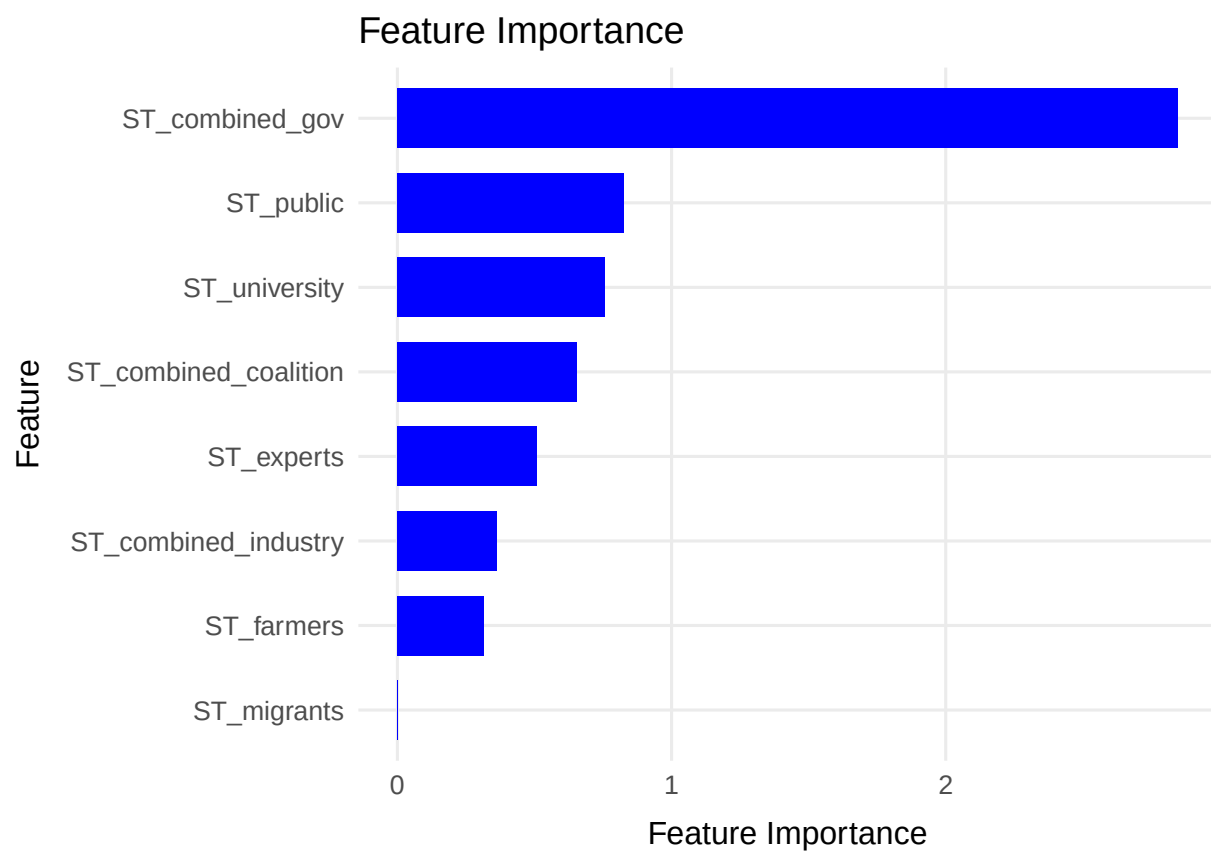
DECISION TREE: Ensembled Decision Tree - stakeholders vs solution

Random forest (ranger). 1000 trees; mtry=2; min.node.size=30; 5-fold CV.

CV AUC: 0.826 (SD=0.113). **n:** 483. **Class counts:** N=465, Y=18.

Top features: ST_combined_gov, ST_public, ST_university.





Researcher Modeling - solution proposed

REGRESSION: Does researcher type increase the likelihood that a solution will be proposed?

Regression of whether researcher type predicts if a solution was proposed (Y/N).

```
##
## Call:
## glm(formula = solution_proposed_YN ~ R_ngo + R_eng + R_math +
##      R_compsci + R_phys + R_interdis + R_socsci + R_economics +
##      R_ag + R_other, family = binomial, data = crcdata)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -3.32515    0.57570  -5.776 7.66e-09 ***
## R_ngo        -14.30039   1500.10090  -0.010  0.9924
## R_eng         -0.60890    0.52621  -1.157  0.2472
## R_math       -13.08984   1340.85939  -0.010  0.9922
## R_compsci     -0.08763    0.66877  -0.131  0.8957
## R_phys        -0.26009    0.50876  -0.511  0.6092
## R_interdis     1.50781    0.53459   2.820  0.0048 **
## R_socsci       -0.38366    0.66670  -0.575  0.5650
## R_economics   -0.98563    1.06464  -0.926  0.3546
## R_ag          -0.37850    0.78165  -0.484  0.6282
## R_other        1.13626    1.13748   0.999  0.3178
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 153.75  on 482  degrees of freedom
## Residual deviance: 137.74  on 472  degrees of freedom
## AIC: 159.74
##
## Number of Fisher Scoring iterations: 16
```

ODDS RATIOS: Does researcher type increase the likelihood that a solution will be proposed?

Odds of whether researcher type predicts if a solution was proposed (Y/N). A ratio which is closer to 1 has a greater level of researcher diversity.

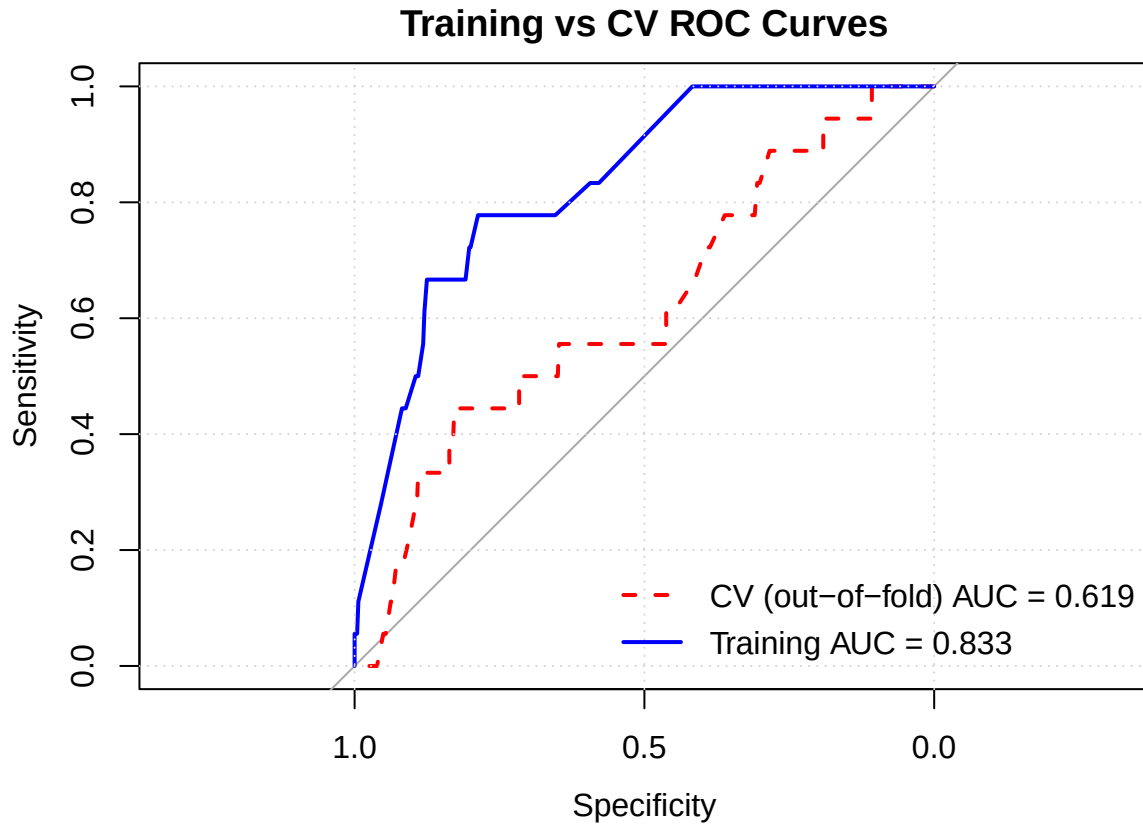
```
##
## Logistic regression predicting solution_proposed_YN : Y vs N
##
##          crude OR(95%CI)    adj. OR(95%CI)    P(Wald's test) P(LR-test)
## Rngo: 1 vs 0      0 (0,Inf)      0 (0,Inf)      0.992      0.481
##
## Reng: 1 vs 0      0.44 (0.17,1.16)  0.54 (0.19,1.53)  0.247      0.242
##
## Rmath: 1 vs 0     0 (0,Inf)      0 (0,Inf)      0.992      0.661
##
## Rcompsci: 1 vs 0  0.93 (0.26,3.3)  0.92 (0.25,3.4)  0.896      0.895
##
## Rphys: 1 vs 0     0.82 (0.32,2.11)  0.77 (0.28,2.09)  0.609      0.608
##
## Rinterdis: 1 vs 0 5.05 (1.86,13.72) 4.52 (1.58,12.88) 0.005      0.004
##
## Rsocsci: 1 vs 0   0.93 (0.26,3.3)  0.68 (0.18,2.52)  0.565      0.552
##
## Reconomics: 1 vs 0 0.33 (0.04,2.49)  0.37 (0.05,3.01)  0.355      0.293
##
## Rag: 1 vs 0       0.5 (0.11,2.21)  0.68 (0.15,3.17)  0.628      0.614
##
## Rother: 1 vs 0    2.22 (0.27,18.07) 3.12 (0.34,28.95) 0.318      0.373
##
## Log-likelihood = -68.8704
## No. of observations = 483
## AIC value = 159.7409
```

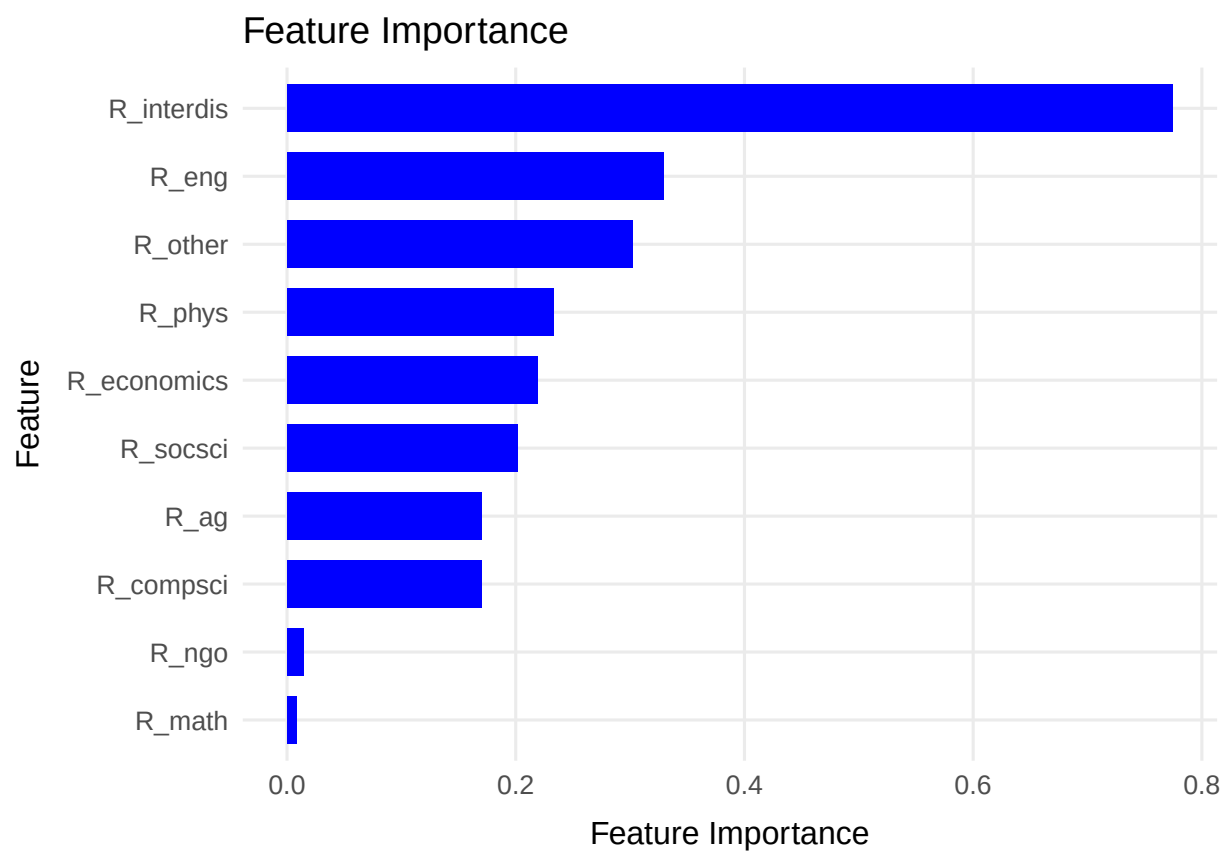

DECISON TREE: Researcher Type Ensembled Decision Tree - researcher type vs solution

Random forest (ranger). 1000 trees; mtry=2; min.node.size=30; 5-fold CV.

CV AUC: 0.635 (SD=0.141). **n:** 483. **Class counts:** N=465, Y=18.

Top features: R_interdis, R_eng, R_other.





Researcher Diversity Modeling - solution proposed

REGRESSION: Does the diversity of researchers increases the likelihood that a solution will be proposed?

Regression of whether Diversity of researchers predicts if a solution was proposed (Y/N). In order to represent diversity, we have used a simple ratio calculation which sums the number of researcher types involved, divided by the total number of possible researcher options. A ratio which is closer to 1 has a greater level of researcher diversity.

```
##
## Call:
## glm(formula = solution_proposed_YN ~ R_ratio, family = binomial,
##      data = crcdata)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -3.1437      0.4846  -6.487 8.78e-11 ***
## R_ratio      -0.5161      2.0451  -0.252   0.801
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 153.75  on 482  degrees of freedom
## Residual deviance: 153.68  on 481  degrees of freedom
## AIC: 157.68
##
## Number of Fisher Scoring iterations: 6
```

ODDS RATIOS: Does the diversity of researchers increases the likelihood that a solution will be proposed?

Odds of whether Diversity of researchers predicts if a solution was proposed (Y/N). In order to represent diversity, we have used a simple ratio calculation which sums the number of researcher types involved, divided by the total number of possible researcher options. A ratio which is closer to 1 has a greater level of researcher diversity.

```
##
## Logistic regression predicting solution_proposed_YN : Y vs N
##
##               OR(95%CI)          P(Wald's test) P(LR-test)
## R_ratio (cont. var.) 0.6 (0.01,32.86)  0.801          0.799
##
## Log-likelihood = -76.8414
## No. of observations = 483
## AIC value = 157.6829
```

Stakeholder Engagement Modeling - Ghodsvali

Regression Testing - Stakeholder type vs level of engagement (Ghodsvali)

```
## Response ST_farmers :
##
## Call:
## lm(formula = ST_farmers ~ STE_G_nominal + STE_G_instrumental +
##     STE_G_representation + STE_G_transformative, data = crcdata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.4444  0.0000  0.0000  0.0000  0.7273
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.324e-16  1.025e-02   0.000      1
## STE_G_nominal    3.929e-01  3.992e-02   9.841 < 2e-16 ***
## STE_G_instrumental 3.684e-01  3.467e-02  10.627 < 2e-16 ***
## STE_G_representation 2.727e-01  6.241e-02   4.370 1.52e-05 ***
## STE_G_transformative 4.444e-01  6.882e-02   6.458 2.62e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2042 on 478 degrees of freedom
## Multiple R-squared:  0.3332, Adjusted R-squared:  0.3276
## F-statistic: 59.71 on 4 and 478 DF,  p-value: < 2.2e-16
##
##
## Response ST_combined_gov :
##
## Call:
## lm(formula = ST_combined_gov ~ STE_G_nominal + STE_G_instrumental +
##     STE_G_representation + STE_G_transformative, data = crcdata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.68421 -0.00252 -0.00252 -0.00252  0.99748
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.002519  0.008868   0.284   0.776
## STE_G_nominal    0.283195  0.034548   8.197 2.28e-15 ***
## STE_G_instrumental 0.681692  0.030003  22.721 < 2e-16 ***
## STE_G_representation 0.997481  0.054006  18.470 < 2e-16 ***
## STE_G_transformative 0.997481  0.059559  16.748 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1767 on 478 degrees of freedom
## Multiple R-squared:  0.6938, Adjusted R-squared:  0.6913
## F-statistic: 270.8 on 4 and 478 DF,  p-value: < 2.2e-16
##
##
```

```

## Response ST_tribal :
##
## Call:
## lm(formula = ST_tribal ~ STE_G_nominal + STE_G_instrumental +
##     STE_G_representation + STE_G_transformative, data = crcdata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.1429  0.0000  0.0000  0.0000  0.9737
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.662e-17  4.816e-03   0.000   1.000
## STE_G_nominal    1.429e-01  1.876e-02   7.613 1.44e-13 ***
## STE_G_instrumental  2.632e-02  1.630e-02   1.615   0.107
## STE_G_representation -1.034e-17  2.933e-02   0.000   1.000
## STE_G_transformative -8.876e-18  3.235e-02   0.000   1.000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.09597 on 478 degrees of freedom
## Multiple R-squared:  0.1103, Adjusted R-squared:  0.1029
## F-statistic: 14.82 on 4 and 478 DF, p-value: 2.004e-11
##
##
## Response ST_combined_coalition :
##
## Call:
## lm(formula = ST_combined_coalition ~ STE_G_nominal + STE_G_instrumental +
##     STE_G_representation + STE_G_transformative, data = crcdata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.66667 -0.00252 -0.00252 -0.00252  0.99748
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.002519   0.009299   0.271  0.78661
## STE_G_nominal    0.104624   0.036230   2.888  0.00406 **
## STE_G_instrumental  0.339586   0.031463  10.793 < 2e-16 ***
## STE_G_representation 0.270208   0.056635   4.771 2.44e-06 ***
## STE_G_transformative 0.664148   0.062459  10.633 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1853 on 478 degrees of freedom
## Multiple R-squared:  0.3329, Adjusted R-squared:  0.3273
## F-statistic: 59.64 on 4 and 478 DF, p-value: < 2.2e-16
##
##
## Response ST_combined_industry :
##
## Call:
## lm(formula = ST_combined_industry ~ STE_G_nominal + STE_G_instrumental +

```

```

##      STE_G_representation + STE_G_transformative, data = crcdata)
##
## Residuals:
##      Min        1Q      Median        3Q        Max
## -0.63636 -0.00252 -0.00252 -0.00252  0.99748
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.002519   0.009921   0.254  0.79969
## STE_G_nominal    0.176053   0.038653   4.555 6.67e-06 ***
## STE_G_instrumental 0.523797   0.033568  15.604 < 2e-16 ***
## STE_G_representation 0.633845   0.060423  10.490 < 2e-16 ***
## STE_G_transformative 0.219703   0.066637   3.297  0.00105 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1977 on 478 degrees of freedom
## Multiple R-squared:  0.4246, Adjusted R-squared:  0.4198
## F-statistic: 88.19 on 4 and 478 DF,  p-value: < 2.2e-16
##
##
## Response ST_migrants :
##
## Call:
## lm(formula = ST_migrants ~ STE_G_nominal + STE_G_instrumental +
##     STE_G_representation + STE_G_transformative, data = crcdata)
##
## Residuals:
##      Min        1Q      Median        3Q        Max
##       0         0         0         0         0
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)           0          0      NaN      NaN
## STE_G_nominal           0          0      NaN      NaN
## STE_G_instrumental      0          0      NaN      NaN
## STE_G_representation     0          0      NaN      NaN
## STE_G_transformative     0          0      NaN      NaN
##
## Residual standard error: 0 on 478 degrees of freedom
## Multiple R-squared:    NaN, Adjusted R-squared:    NaN
## F-statistic:    NaN on 4 and 478 DF,  p-value: NA
##
##
## Response ST_youth :
##
## Call:
## lm(formula = ST_youth ~ STE_G_nominal + STE_G_instrumental +
##     STE_G_representation + STE_G_transformative, data = crcdata)
##
## Residuals:
##      Min        1Q      Median        3Q        Max
## -0.05263 -0.00252 -0.00252 -0.00252  0.99748
##

```

```

## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      0.002519   0.004508   0.559  0.57659
## STE_G_nominal      0.033195   0.017563   1.890  0.05936 .
## STE_G_instrumental  0.050113   0.015252   3.286  0.00109 **
## STE_G_representation -0.002519   0.027455  -0.092  0.92694
## STE_G_transformative -0.002519   0.030278  -0.083  0.93373
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.08982 on 478 degrees of freedom
## Multiple R-squared:  0.02782, Adjusted R-squared:  0.01969
## F-statistic:  3.42 on 4 and 478 DF, p-value: 0.009011
##
##
## Response ST_public :
##
## Call:
## lm(formula = ST_public ~ STE_G_nominal + STE_G_instrumental +
##     STE_G_representation + STE_G_transformative, data = crcdata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.2857   0.0000   0.0000   0.0000   0.8684
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -3.536e-17  8.355e-03   0.000 1.000000
## STE_G_nominal     2.857e-01  3.255e-02   8.777 < 2e-16 ***
## STE_G_instrumental  1.316e-01  2.827e-02   4.654 4.21e-06 ***
## STE_G_representation  1.818e-01  5.089e-02   3.573 0.000389 ***
## STE_G_transformative  2.222e-01  5.612e-02   3.960 8.64e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1665 on 478 degrees of freedom
## Multiple R-squared:  0.1923, Adjusted R-squared:  0.1855
## F-statistic: 28.44 on 4 and 478 DF, p-value: < 2.2e-16
##
##
## Response ST_university :
##
## Call:
## lm(formula = ST_university ~ STE_G_nominal + STE_G_instrumental +
##     STE_G_representation + STE_G_transformative, data = crcdata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.55556 -0.00252 -0.00252 -0.00252  0.99748
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      0.002519   0.009911   0.254 0.799480
## STE_G_nominal      0.140338   0.038611   3.635 0.000309 ***

```



```

## STE_G_instrumental    0.418534    0.033531   12.482 < 2e-16 ***
## STE_G_representation 0.452027    0.060358    7.489 3.37e-13 ***
## STE_G_transformative 0.553037    0.066564    8.308 1.01e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1975 on 478 degrees of freedom
## Multiple R-squared:  0.3575, Adjusted R-squared:  0.3521
## F-statistic: 66.5 on 4 and 478 DF,  p-value: < 2.2e-16
##
##
## Response ST_experts :
##
## Call:
## lm(formula = ST_experts ~ STE_G_nominal + STE_G_instrumental +
##     STE_G_representation + STE_G_transformative, data = crcdata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.28947 -0.00252 -0.00252 -0.00252  0.99748
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.002519   0.009180   0.274   0.7839
## STE_G_nominal    0.176053   0.035764   4.923 1.18e-06 ***
## STE_G_instrumental 0.286955   0.031059   9.239 < 2e-16 ***
## STE_G_representation 0.270208   0.055907   4.833 1.81e-06 ***
## STE_G_transformative 0.108592   0.061656   1.761  0.0788 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1829 on 478 degrees of freedom
## Multiple R-squared:  0.2039, Adjusted R-squared:  0.1972
## F-statistic: 30.61 on 4 and 478 DF,  p-value: < 2.2e-16

```

Regression Testing - Stakeholder type vs solution

```
##
## Call:
## glm(formula = solution_proposed_YN ~ ST_farmers + ST_combined_gov +
##      ST_tribal + ST_combined_coalition + ST_combined_industry +
##      ST_migrants + ST_youth + ST_public + ST_university + ST_experts,
##      family = binomial, data = crcdata)
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -4.67323    0.50476  -9.258  < 2e-16 ***
## ST_farmers      0.56210    0.72644   0.774    0.439
## ST_combined_gov  3.96372    0.80612   4.917 8.79e-07 ***
## ST_tribal     -14.36229 1623.97493  -0.009    0.993
## ST_combined_coalition -0.01271    0.87795  -0.014    0.988
## ST_combined_industry -0.97124    0.74281  -1.308    0.191
## ST_migrants           NA          NA      NA      NA
## ST_youth       -15.26565 1784.23408  -0.009    0.993
## ST_public        0.74554    0.79355   0.940    0.347
## ST_university   -0.23106    0.82786  -0.279    0.780
## ST_experts      -0.29851    0.80371  -0.371    0.710
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 153.75  on 482  degrees of freedom
## Residual deviance: 102.25  on 473  degrees of freedom
## AIC: 122.25
##
## Number of Fisher Scoring iterations: 16
```

Geographic Location Modeling - solution proposed

REGRESSION: Does the geographic location of the study increase the likelihood of proposed/implemented solutions?

```
##
## Call:
## glm(formula = solution_proposed_YN ~ G_local + G_regional + G_national +
##       G_multinational + G_global, family = binomial, data = crcdata)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -3.6437     0.9147  -3.984 6.79e-05 ***
## G_local         0.6433     0.9908   0.649   0.516
## G_regional     0.1523     1.0042   0.152   0.879
## G_national     1.0220     1.0071   1.015   0.310
## G_multinational -14.9224  1232.6632  -0.012   0.990
## G_global      -14.9224  1966.6497  -0.008   0.994
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 153.75  on 482  degrees of freedom
## Residual deviance: 148.27  on 477  degrees of freedom
## AIC: 160.27
##
## Number of Fisher Scoring iterations: 17
```

ODDS RATIOS: Does the geographic location of the study increase the likelihood of proposed/implemented solutions?

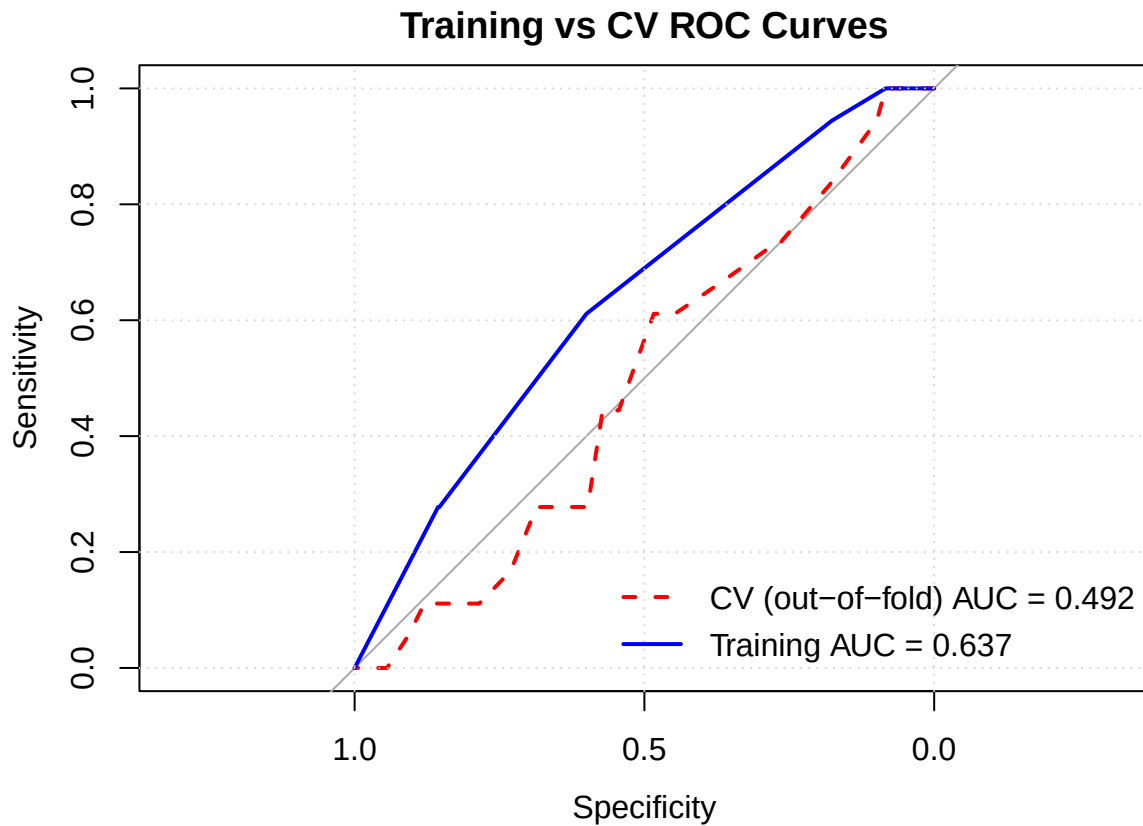
```
##
## Logistic regression predicting solution_proposed_YN : Y vs N
##
##          crude OR(95%CI)   adj. OR(95%CI)   P(Wald's test) P(LR-test)
## G_local: 1 vs 0          1.45 (0.53,3.96)   1.9 (0.27,13.27)  0.516          0.494
##
## G_regional: 1 vs 0       0.68 (0.25,1.84)   1.16 (0.16,8.33)  0.879          0.878
##
## G_national: 1 vs 0       2.25 (0.78,6.5)    2.78 (0.39,20)    0.31           0.277
##
## G_multinational: 1 vs 0  0 (0,Inf)          0 (0,Inf)          0.99           0.285
##
## G_global: 1 vs 0         0 (0,Inf)          0 (0,Inf)          0.994          0.475
##
## Log-likelihood = -74.1333
## No. of observations = 483
## AIC value = 160.2665
```

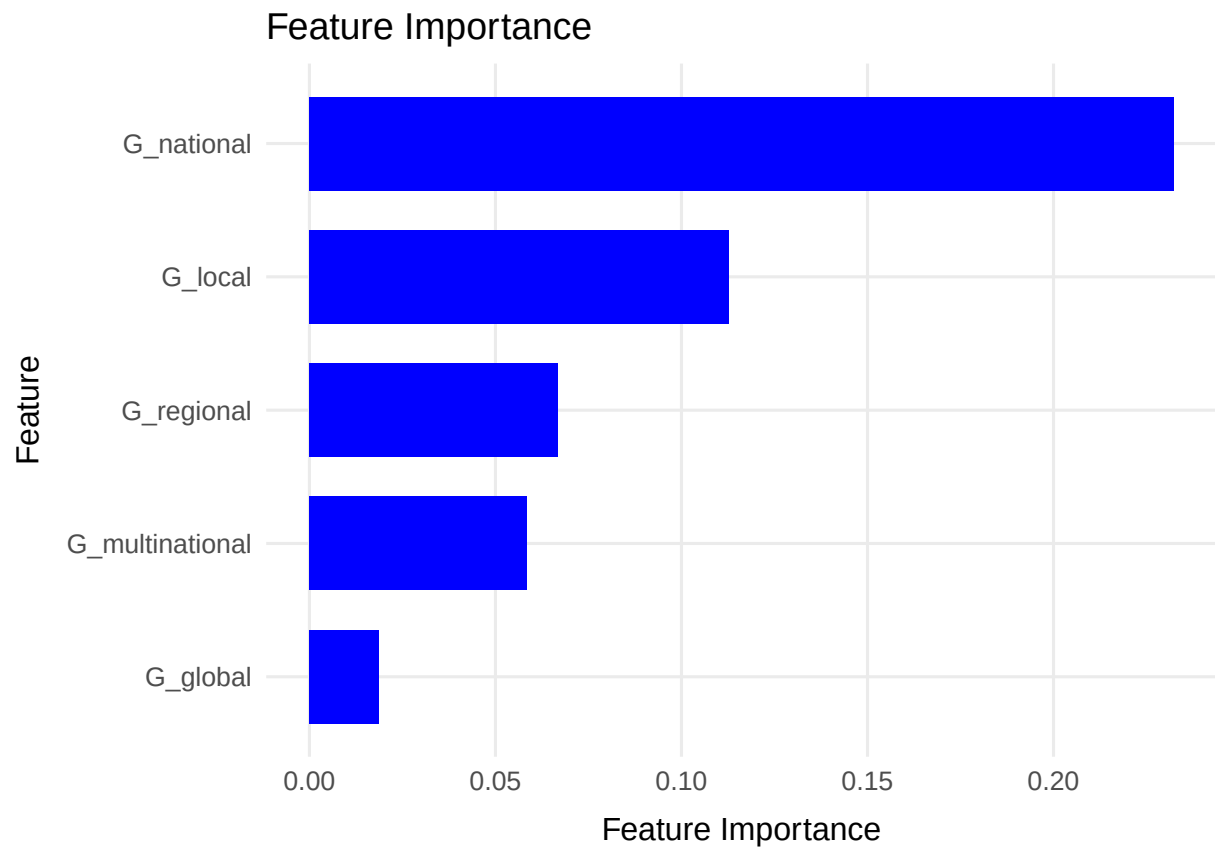
DECISON TREE: Geographic area Ensembled Decision Tree - Geographic area vs solution

Random forest (ranger). 1000 trees; mtry=2; min.node.size=20; 5-fold CV.

CV AUC: 0.492 (SD=0.124). **n:** 483. **Class counts:** N=465, Y=18.

Top features: G_national, G_local, G_regional.





Regional Location Modeling - solution proposed

REGRESSION: Does the regional location of the study increase the likelihood of proposed/implemented solutions?

Regions were grouped in: Europe/Asia, Middle East/Global - and Other. Results of this bias-reduced logistic regression indicate that region was significantly associated with whether a solution was proposed, Chisquare = 7.28, p = .026.

```
##
## Call:
## glm(formula = solution_proposed_YN ~ L_region_3, family = binomial,
##      data = crcdata, method = "brglmFit")
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -0.4009  -0.3788  -0.1930  -0.1930   2.8257
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -3.9737     0.4311  -9.217  < 2e-16 ***
## L_region_3EuropeAsia  1.4932     0.5485   2.722  0.00648 **
## L_region_3GlobalME    1.3752     0.6545   2.101  0.03563 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 153.76  on 482  degrees of freedom
## Residual deviance: 145.04  on 480  degrees of freedom
## AIC: 151.04
##
## Type of estimator: AS_mixed (mixed bias-reducing adjusted score equations)
## Number of Fisher Scoring iterations: 3
```

ODDS RATIOS: Does the regional location of the study increase the likelihood of proposed/implemented solutions?

Compared to other regions, cases from Europe and Asia had 3.77 times higher odds of proposing a solution (95% CI [1.35, 10.48], $p = .011$). Global and Middle East cases also had higher odds (OR = 3.35, 95% CI [0.97, 11.57]), though this effect was marginal ($p = .056$).

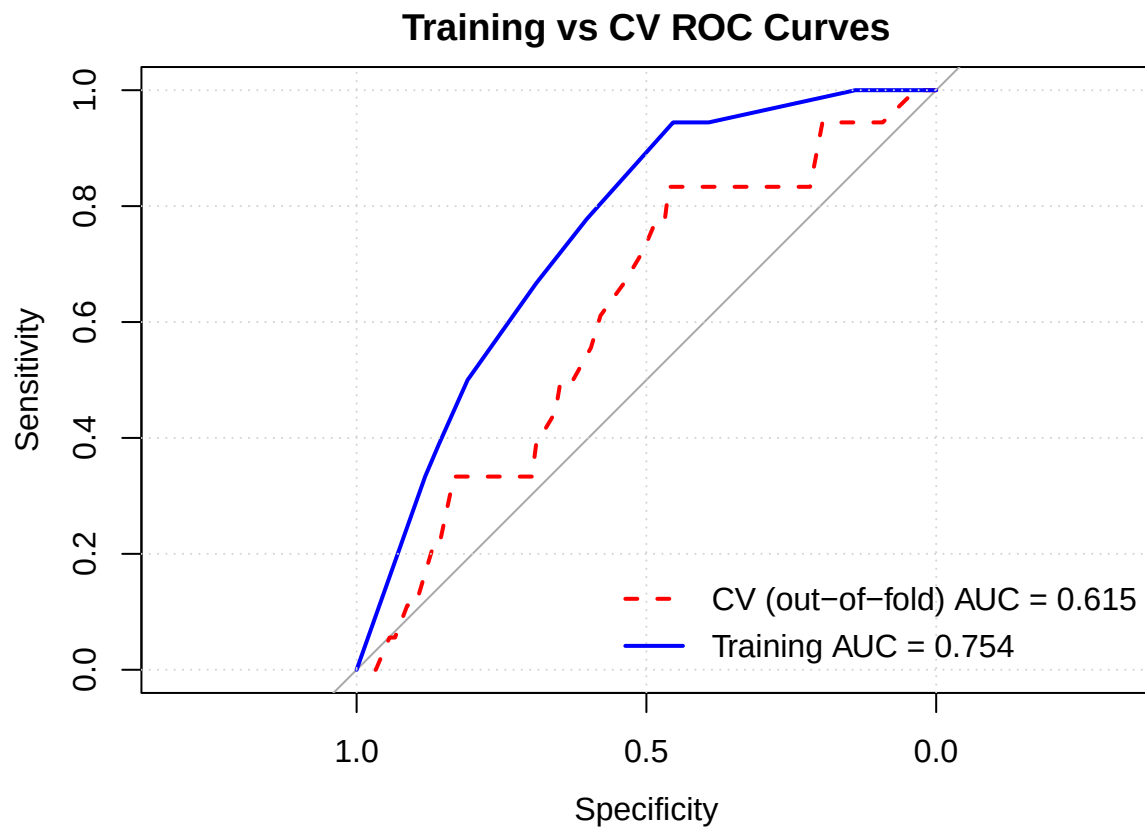
##	OR	2.5 %	97.5 %
## (Intercept)	0.01880342	0.008077381	0.04377268
## L_region_3EuropeAsia	4.45134158	1.519125103	13.04332462
## L_region_3GlobalME	3.95567490	1.096799990	14.26637862

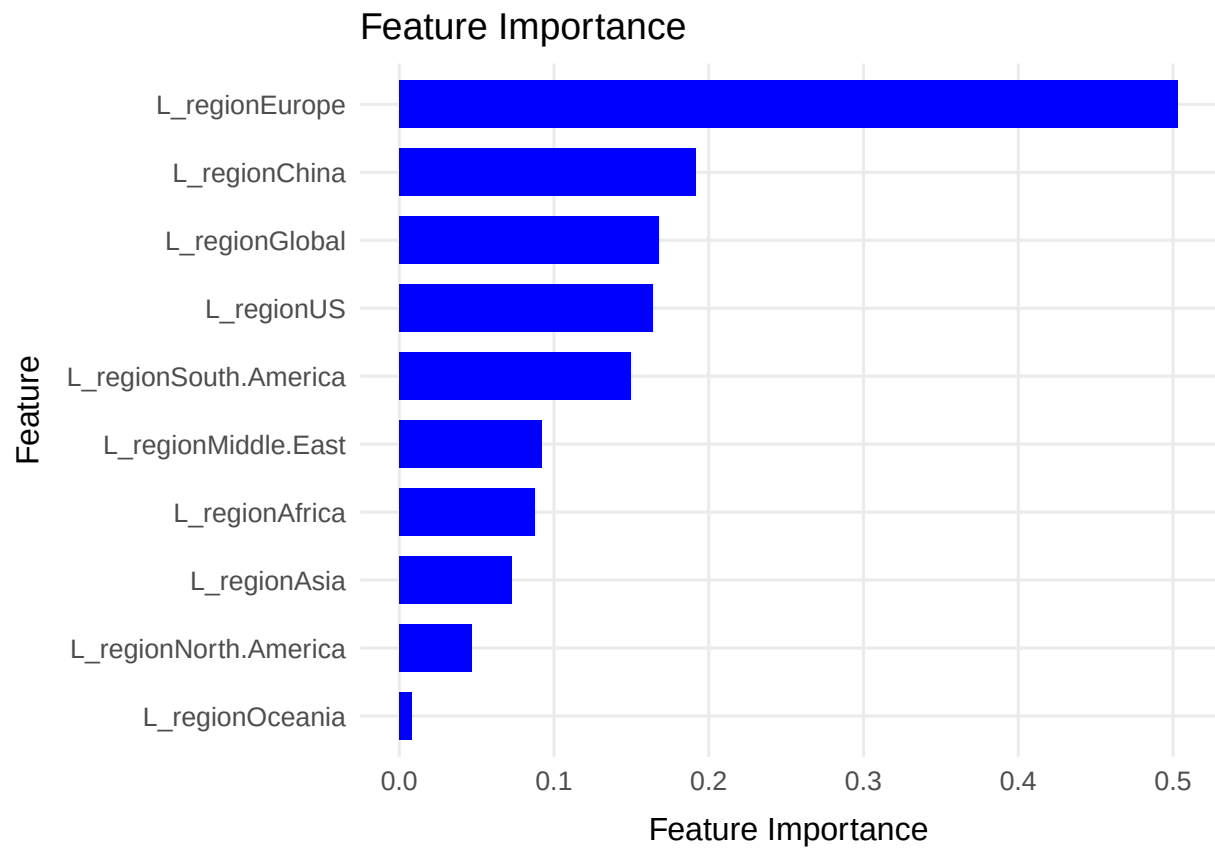
DECISON TREE: Region area Ensembded Decision Tree - Region area vs solution

Random forest (ranger). 1000 trees; mtry=2; min.node.size=30; 5-fold CV.

CV AUC: 0.635 (SD=0.120). **n:** 441. **Class counts:** N=423, Y=18.

Top features: L_regionEurope, L_regionChina, L_regionGlobal.





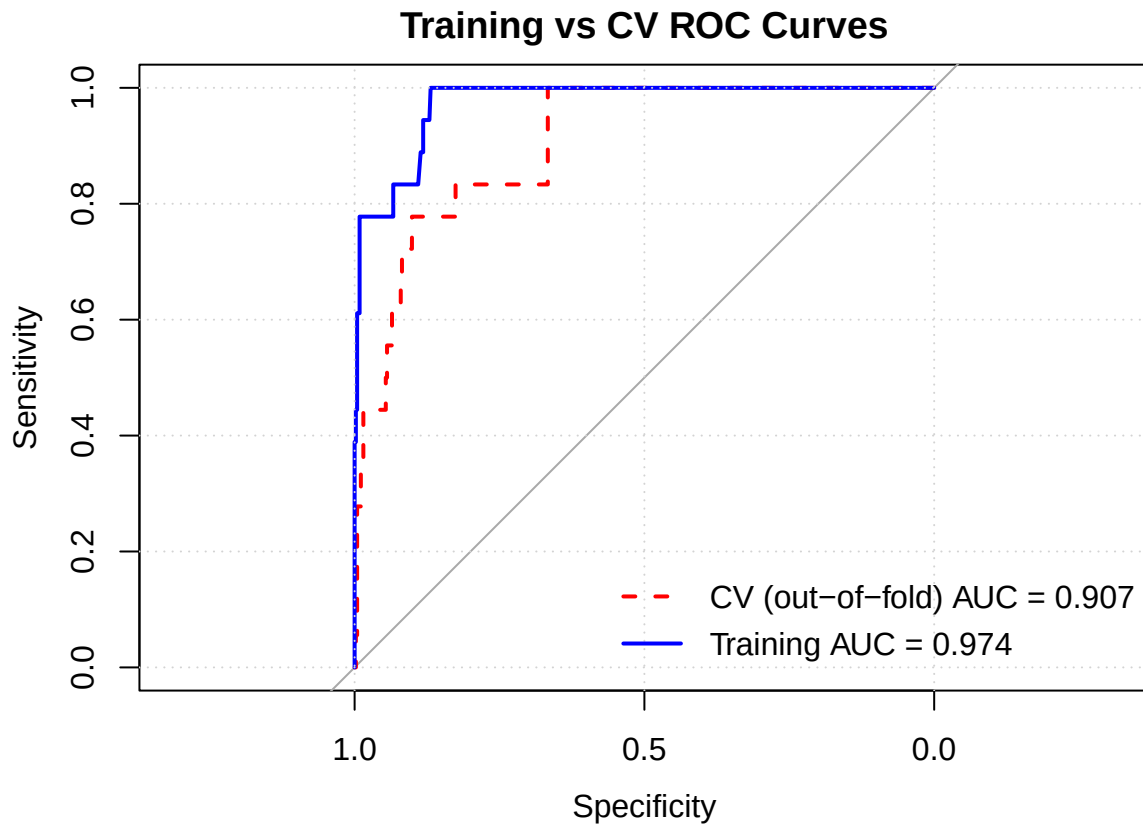
DECISION TREE ANALYSIS - ALL VARIABLES

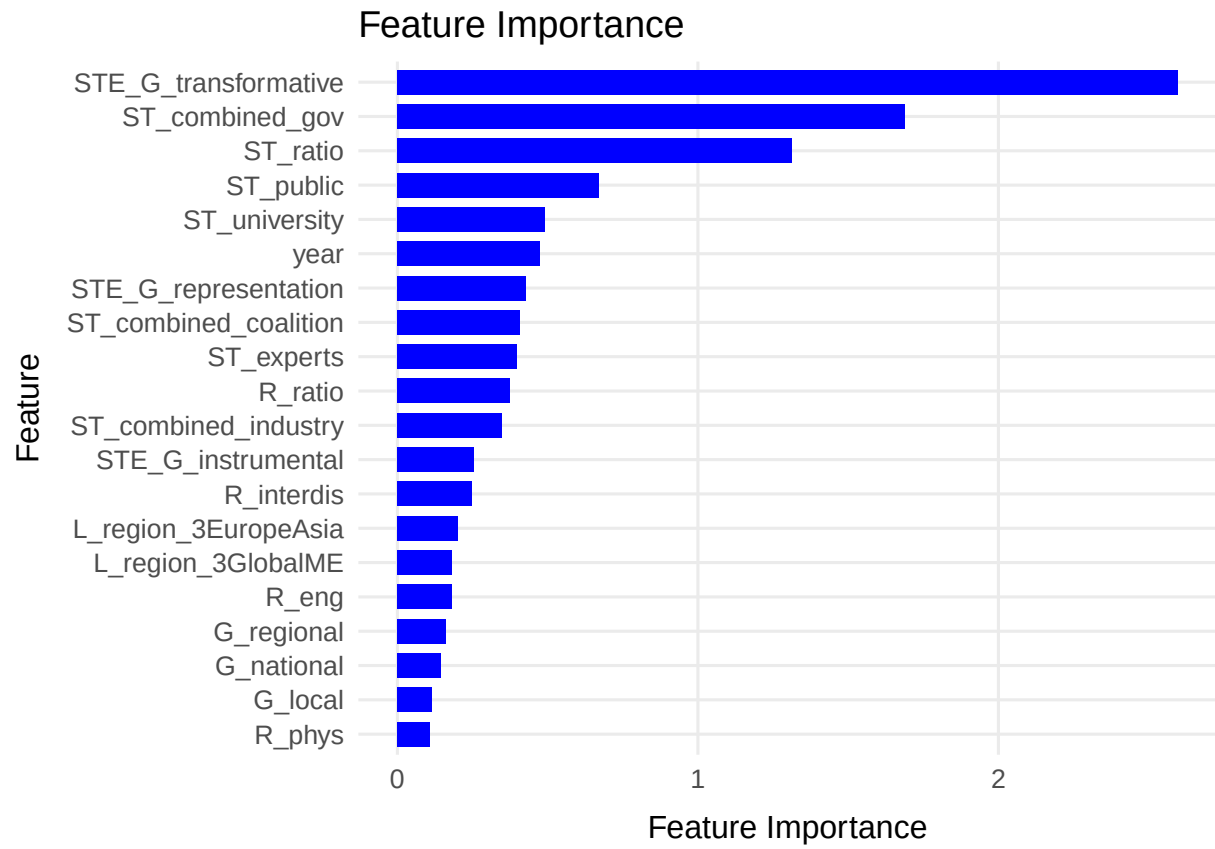
Looking at Decision Tree for all variables, including the Ghodsvali scale - with solution proposed as dependent variable

```
## **Random forest (ranger).** 1000 trees; mtry=2; min.node.size=20; 5-fold CV.
```

```
## **CV AUC:** 0.930 (SD=0.066). **n:** 483. **Class counts:** N=465, Y=18.
```

```
## **Top features:** STE_G_transformative, ST_combined_gov, ST_ratio.
```





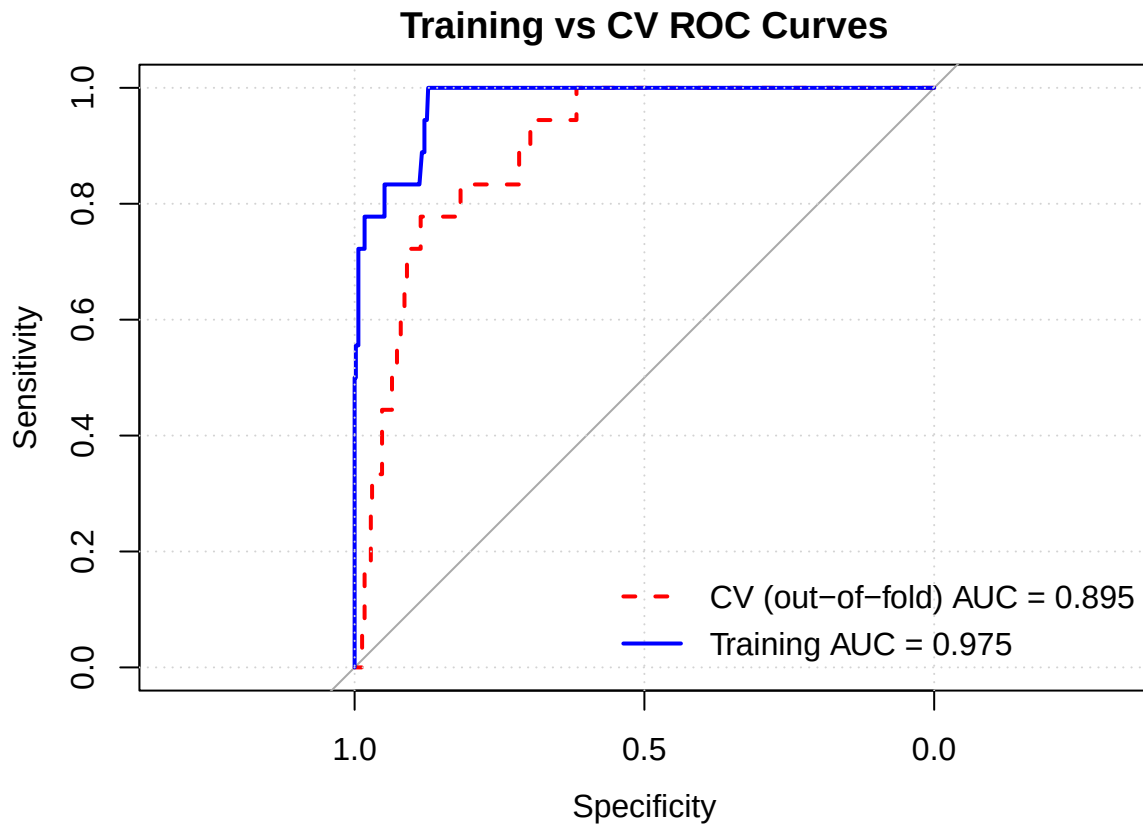
DECISION TREE ANALYSIS - ALL VARIABLES - minus scaling

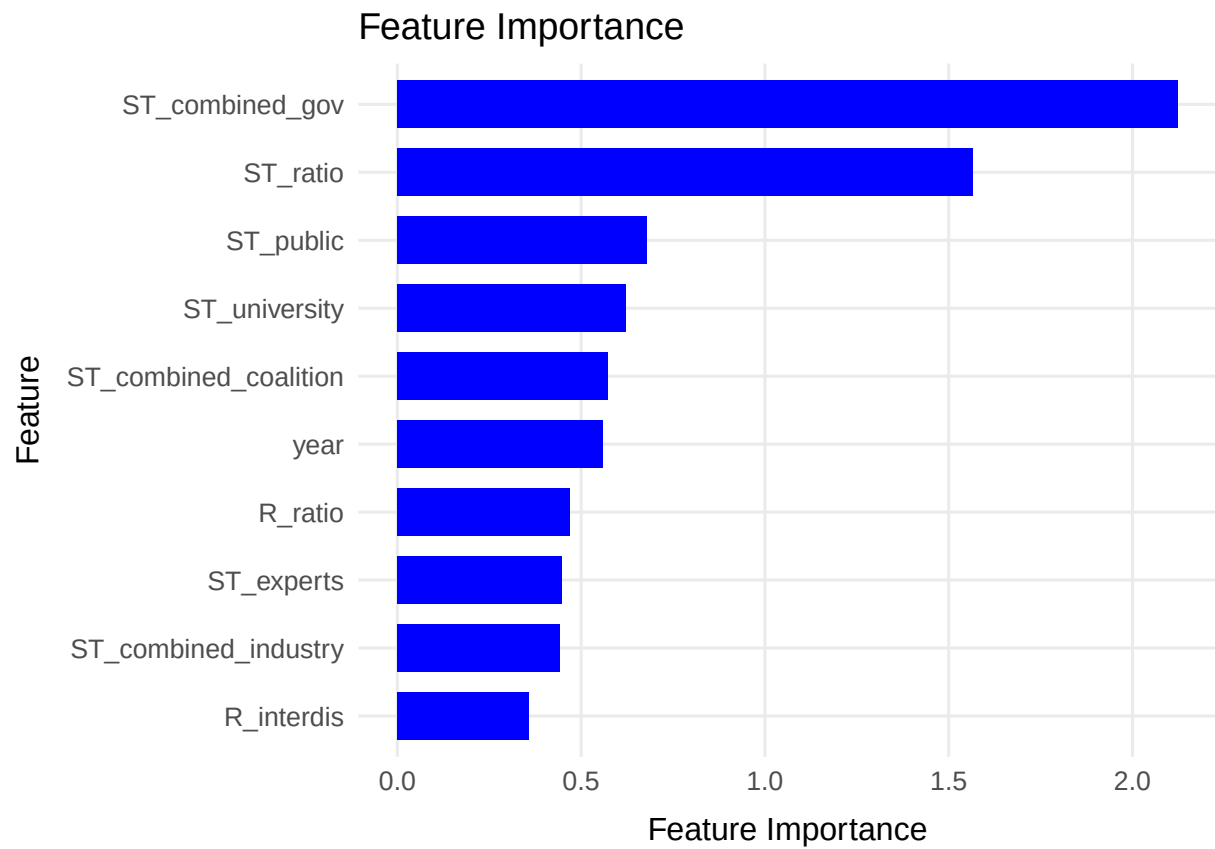
Looking at Decision Tree for all variables - minus the Ghodsvali scale - with solution proposed as dependent variable

```
## **Random forest (ranger).** 1000 trees; mtry=2; min.node.size=20; 5-fold CV.
```

```
## **CV AUC:** 0.909 (SD=0.065). **n:** 483. **Class counts:** N=465, Y=18.
```

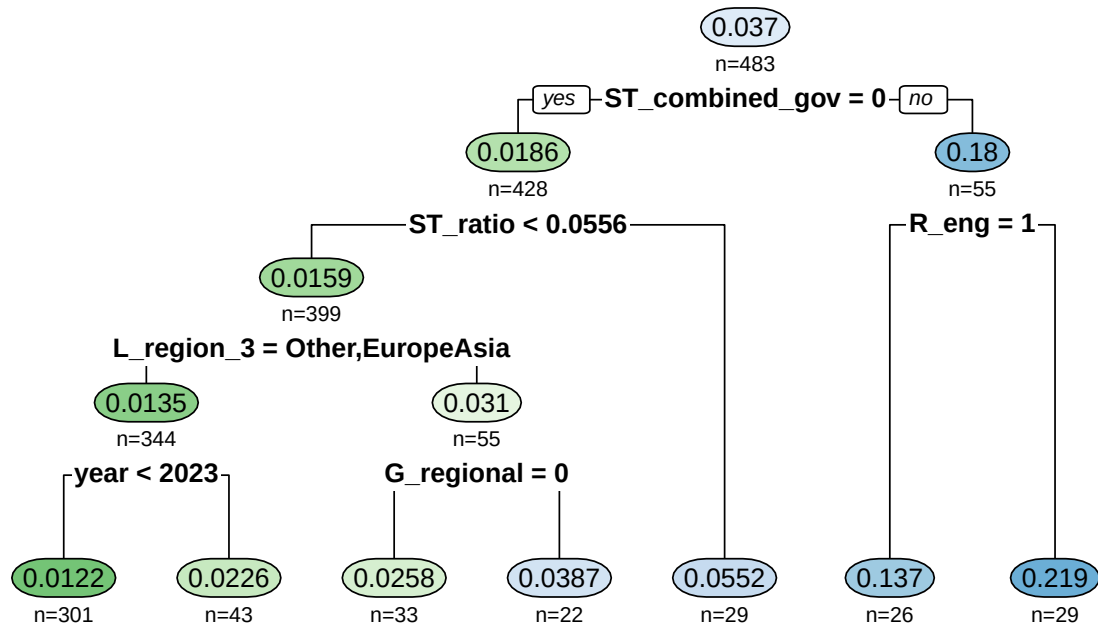
```
## **Top features:** ST_combined_gov, ST_ratio, ST_public.
```





Representative Decision Tree Plot - Balanced Model - Minus Scaling

Representative Tree



[1] 0.0004966553

Representative Decision Tree Plot - All variables

pdf
2

Representative Tree

